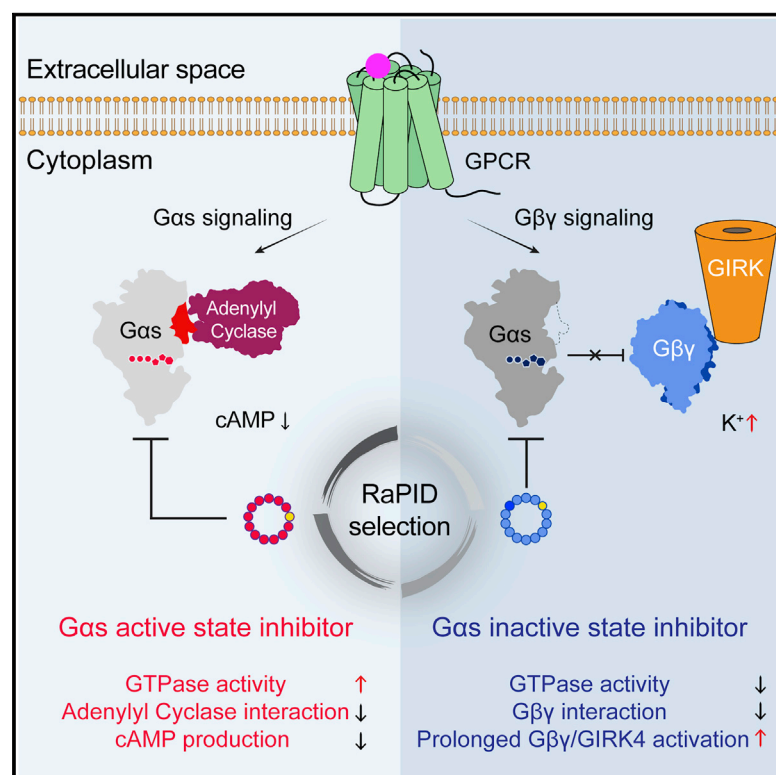


State-selective modulation of heterotrimeric $G_{\alpha s}$ signaling with macrocyclic peptides

Graphical abstract



Authors

Shizhong A. Dai, Qi Hu, Rong Gao, ..., Mark von Zastrow, Hiroaki Suga, Kevan M. Shokat

Correspondence

hsuga@chem.s.u-tokyo.ac.jp (H.S.), kevan.shokat@ucsf.edu (K.M.S.)

In brief

Cell-permeable $G_{\alpha s}$ inhibitors regulate G protein signal transduction with high nucleotide-binding-state selectivity and class specificity.

Highlights

- Discovery of nucleotide-state-selective cyclic peptide binders for $G_{\alpha s}$
- Co-crystal structures reveal G protein class specificity of the cyclic peptides
- Inhibition of the $G_{\alpha s}$ ON-state reduces isoproterenol-stimulated cAMP production
- Sequestering the $G_{\alpha s}$ OFF-state prolongs $G\beta\gamma$ activation upon receptor activation



Article

State-selective modulation of heterotrimeric G α s signaling with macrocyclic peptides

Shizhong A. Dai,^{1,2,8} Qi Hu,^{1,2,6,8} Rong Gao,^{3,8} Emily E. Blythe,^{1,4} Kouki K. Touhara,⁵ Hayden Peacock,³ Ziyang Zhang,^{1,2,7} Mark von Zastrow,^{1,4} Hiroaki Suga,^{3,*} and Kevan M. Shokat^{1,2,9,*}

¹Department of Cellular and Molecular Pharmacology, University of California, San Francisco, San Francisco, CA 94158, USA

²Howard Hughes Medical Institute, Chevy Chase, MD 20815, USA

³Department of Chemistry, Graduate School of Science, The University of Tokyo, 7-3-1 Hongo, Bunkyo-ku, Tokyo 113-0033, Japan

⁴Department of Psychiatry, University of California, San Francisco, San Francisco, CA 94158, USA

⁵Department of Physiology, University of California, San Francisco, San Francisco, CA 94158, USA

⁶Present address: School of Life Sciences, Westlake University, Hangzhou, Zhejiang, China

⁷Present address: Department of Chemistry, University of California, Berkeley, Berkeley, CA, 94720, USA

⁸These authors contributed equally

⁹Lead contact

*Correspondence: hsuga@chem.s.u-tokyo.ac.jp (H.S.), kevan.shokat@ucsf.edu (K.M.S.)

<https://doi.org/10.1016/j.cell.2022.09.019>

SUMMARY

The G protein-coupled receptor cascade leading to production of the second messenger cAMP is replete with pharmacologically targetable proteins, with the exception of the G α subunit, G α s. GTPases remain largely undruggable given the difficulty of displacing high-affinity guanine nucleotides and the lack of other drug binding sites. We explored a chemical library of 10¹² cyclic peptides to expand the chemical search for inhibitors of this enzyme class. We identified two macrocyclic peptides, GN13 and GD20, that antagonize the active and inactive states of G α s, respectively. Both macrocyclic peptides fine-tune G α s activity with high nucleotide-binding-state selectivity and G protein class-specificity. Co-crystal structures reveal that GN13 and GD20 distinguish the conformational differences within the switch II/ α 3 pocket. Cell-permeable analogs of GN13 and GD20 modulate G α s/G $\beta\gamma$ signaling in cells through binding to crystallographically defined pockets. The discovery of cyclic peptide inhibitors targeting G α s provides a path for further development of state-dependent GTPase inhibitors.

INTRODUCTION

The family of human GTPases represents a vast but largely untapped source of pharmacological targets. They serve as key molecular switches that control cell growth and proliferation through cycling between tightly regulated ON/OFF states. The role of specific GTPase family members across diverse human diseases has been widely established by cancer genome sequencing (e.g., *KRAS* and *GNAS*) and by familial studies in neurodegenerative disease (e.g., *LRRK2* and *RAB39B*) (Prior et al., 2012; O'Hayre et al., 2013; Alessi and Sammler, 2018; Wilson et al., 2014). Despite the widespread recognition of these disease target relationships, only very recently has the first drug targeting a GTPase K-Ras(G12C) achieved clinical proof of principle (Canon et al., 2019; Hallin et al., 2020) by covalently targeting a somatic mutant cysteine.

Several peptide-based probes that non-covalently target GTPases have been reported, but they either lack proper drug-like properties or have limited target scope (Takasaki et al., 2004; Ja and Roberts, 2004; Johnston et al., 2005; Johnston et al., 2005; Johnston et al., 2006; Ja et al., 2006; Austin et al., 2008). Short linear peptides have been shown to state-selectively

target the switch II/ α 3 pocket in the heterotrimeric G protein α -subunit (G α). However, linear peptides are not the ideal molecules for drug discovery because of their poor cell permeability and instability in cells.

Cyclic peptides are promising candidates for GTPase drug development. Like linear peptides, cyclic peptides are also capable of targeting protein-protein interfaces (Sohrabi et al., 2020). Peptide cyclization stabilizes the peptide sequence and constrains the peptide conformation for better cell penetration (Dougherty et al., 2019). Cyclic peptide inhibitors of G α proteins have been reported; for instance, the cyclic depsipeptide natural product YM-254890 targets GDP-bound G α q with high specificity and potency (Nishimura et al., 2010). Despite the highly conserved structure of G proteins and the recent total synthesis of YM-254890, efforts to use this macrocycle as a scaffold from which to discover inhibitors of other G proteins (G α s, G α i) have not been successful (Kaur et al., 2015; Xiong et al., 2016; Zhang et al., 2017), likely due to the limited chemical diversity of available YM-254890 analogs. We therefore reasoned that screening an ultra-large library of cyclic peptides against a given nucleotide binding state of G α s might allow us to discover G α s inhibitors that discriminate between the active and inactive states of G α s.



and potentially open the remainder of the GTPase family to pharmacological studies.

The Random nonstandard Peptide Integrated Discovery (RaPID) system (Yamagishi et al., 2011) merges the flexibility of an *in vitro* translation system (Murakami et al., 2003, 2006; Ramaswamy et al., 2004; Xiao et al., 2008) with mRNA display, enabling the screening of exceptionally large macrocyclic peptide libraries ($>10^{12}$ molecules) against challenging targets (Pasioura and Suga, 2017). Here, we report the discovery by the RaPID system of two macrocyclic peptides, GN13 and GD20, and their analogs cpGN13 and cpGD20, that are cell-permeable, nucleotide-state-selective inhibitors of $G_{\alpha s}$, with high selectivity over other G protein subfamilies.

RESULTS

Selection of cyclic peptides that bind to the active or inactive state of $G_{\alpha s}$

The RaPID cyclic peptide discovery platform selects for high-affinity cyclic peptide binders, but hits might bind $G_{\alpha s}$ anywhere on its surface and so might or might not perturb its function. To increase the probability of selecting function-perturbing hits, we took advantage of the fact that when $G_{\alpha s}$ switches from the GDP-bound inactive state to the GTP-bound active state, significant conformational changes occur at the switch I, II, and III regions (Lambright et al., 1994), which are known to bind protein partners such as $G\beta\gamma$ or adenylyl cyclases (AC) (Liu et al., 2019; Tesmer et al., 1997) (Figure 1A). We reasoned that performing a positive selection against one state of $G_{\alpha s}$ and a negative selection against the other state would enrich for binders to the switch regions, and that these binders would be likely to state-selectively disrupt $G_{\alpha s}$ function.

To select $G_{\alpha s}$ active-state binders, we performed a positive selection with wild-type (WT) $G_{\alpha s}$ bound to the non-hydrolyzable GTP analog GppNHP (5'-guanylyl imidodiphosphate [GNP]) and a negative selection against GDP-bound WT $G_{\alpha s}$. A parallel $G_{\alpha s}$ inactive-state binder selection was performed using GDP-bound WT $G_{\alpha s}$ as the positive selection and GNP-bound WT $G_{\alpha s}$ as the negative selection (Figure 1B). There are short and long isoforms of $G_{\alpha s}$, which are splice variants that differ from each other in the hinge region between the Ras domain and the helical domain (Seifert et al., 1998). If not otherwise noted, the short isoform was used in our study.

Starting from a cDNA library, each round of selection included PCR amplification of the cDNA library, *in vitro* transcription into an mRNA library, ligation with a puromycin linker, and translation to generate a peptide library covalently conjugated to their encoding mRNA library (Figure 1C). The library peptides contain an N-chloroacetyl-D-tyrosine at the N terminus, followed by 8–12 random proteinogenic amino acids encoded by NNK codons (N = G, C, A or U; K = G or U), a cysteine residue, and a GSGSGS linker (G = glycine; S = serine, Figures 1D and 1E). Cyclization occurs spontaneously between the chloroacetyl group and the thiol group of the downstream cysteine residue. The peptide-ligated mRNA library was further reverse-transcribed into a cDNA-mRNA-peptide library, subjected to a negative selection against one state of $G_{\alpha s}$, then followed by a positive selection against the other state of $G_{\alpha s}$ (Figure 1C).

After four rounds of selection (R1–R4), cyclic peptide binders for $G_{\alpha s}$ /GNP or $G_{\alpha s}$ /GDP were enriched (Figures S1A and S1B) and identified by next generation sequencing (NGS). The sequences of the top 20 hits are shown in Figures 1D and 1E. Selective cyclic peptides from the R4 pool were characterized by comparison selection against the respective positive and negative protein baits (Figures 1F and 1G, see also Figure S1C). Nine of the top 20 hits from the active-state binder selection (with >100 -fold selectivity for $G_{\alpha s}$ /GNP over $G_{\alpha s}$ /GDP, red triangles in Figure 1D) and eight of the top 20 hits from the inactive-state binder selection (with >40 -fold selectivity for $G_{\alpha s}$ /GDP over $G_{\alpha s}$ /GNP, blue triangles in Figure 1E) were chosen for further analysis. To evaluate the cyclic peptide hits without the appended DNA/mRNA duplex, residues from N-chloroacetyl-D-tyrosine to glycine (after the anchor cysteine residue) of the selected peptides were chemically synthesized, followed by cyclization.

Active-state binding cyclic peptide GN13 blocks $G_{\alpha s}$ -mediated AC activation

To determine whether active-state binders inhibit $G_{\alpha s}$ activity, we assayed the ability of $G_{\alpha s}$ to activate its effector, AC (Figure 2A). We refer to resynthesized active-state binders with a “GN” ($G_{\alpha s}$ /GNP) preceding their ranking number. We first tested the interaction between $G_{\alpha s}$ /GNP and AC in the presence of active-state binders using a fluorescence resonance energy transfer (FRET) assay (Figure S2A). Eight out of nine GN peptides potentially inhibited $G_{\alpha s}$ /AC interaction (Figure 2B). We then performed a reconstituted AC activity assay to test the ability of GN peptides to inhibit $G_{\alpha s}$ -mediated AC activation (Figure 2A). GN13 was the most potent among the top hits, with an IC_{50} of $4.15 \pm 1.13 \mu M$ (Figures 2C and 2D). GN13 did not inhibit $G_{\alpha s}$ -independent AC activity (Figure S2B), suggesting a $G_{\alpha s}$ -dependent mechanism of inhibition. We measured the binding of GN13 to immobilized $G_{\alpha s}$ /GNP using biolayer interferometry (BLI). GN13 binds to $G_{\alpha s}$ /GNP with a K_D value of $0.19 \pm 0.02 \mu M$ (Figure S2C, see also Table S3). By contrast, GN13 showed little to no detectable binding to GDP-bound $G_{\alpha s}$ (Figure S2D).

We next tested the ability of GN13 to inhibit β_2 -adrenergic receptor (β_2 AR)-mediated cAMP production. Membrane anchored GDP-bound $G_{\alpha s}$ forms a heterotrimer with $G\beta\gamma$ in the resting state. Upon agonist stimulation, β_2 AR activates $G_{\alpha s}$ by promoting GDP to GTP exchange (Weis and Kobilka, 2018). We hypothesized that GN13 might capture newly generated GTP-bound $G_{\alpha s}$ and prevent it from binding AC (Figure 2E). We incubated live HEK293 cells or HEK293 cell membranes with GN13 and measured cAMP accumulation with or without β_2 AR stimulation by isoproterenol (ISO). Although GN13 showed no inhibition in live HEK293 cells, it inhibited ISO-stimulated cAMP accumulation in cell membranes to a background level, with an IC_{50} of $12.21 \pm 2.51 \mu M$ (Figures S2E and 2F).

The lack of cell activity of GN13 is presumably limited by its poor cell permeability. We sought to improve its permeability by substituting the negatively charged GN13 with a glutamine residue. We evaluated the cell permeability of GN13-E3Q with a chloroalkane penetration assay (CAPA) (Peraro et al., 2018) (Figure S2F). HeLa cells expressing HaloTag localized to the mitochondrial outer membrane were pulsed with chloroalkane-tagged molecules (ct-molecule), washed, chased with

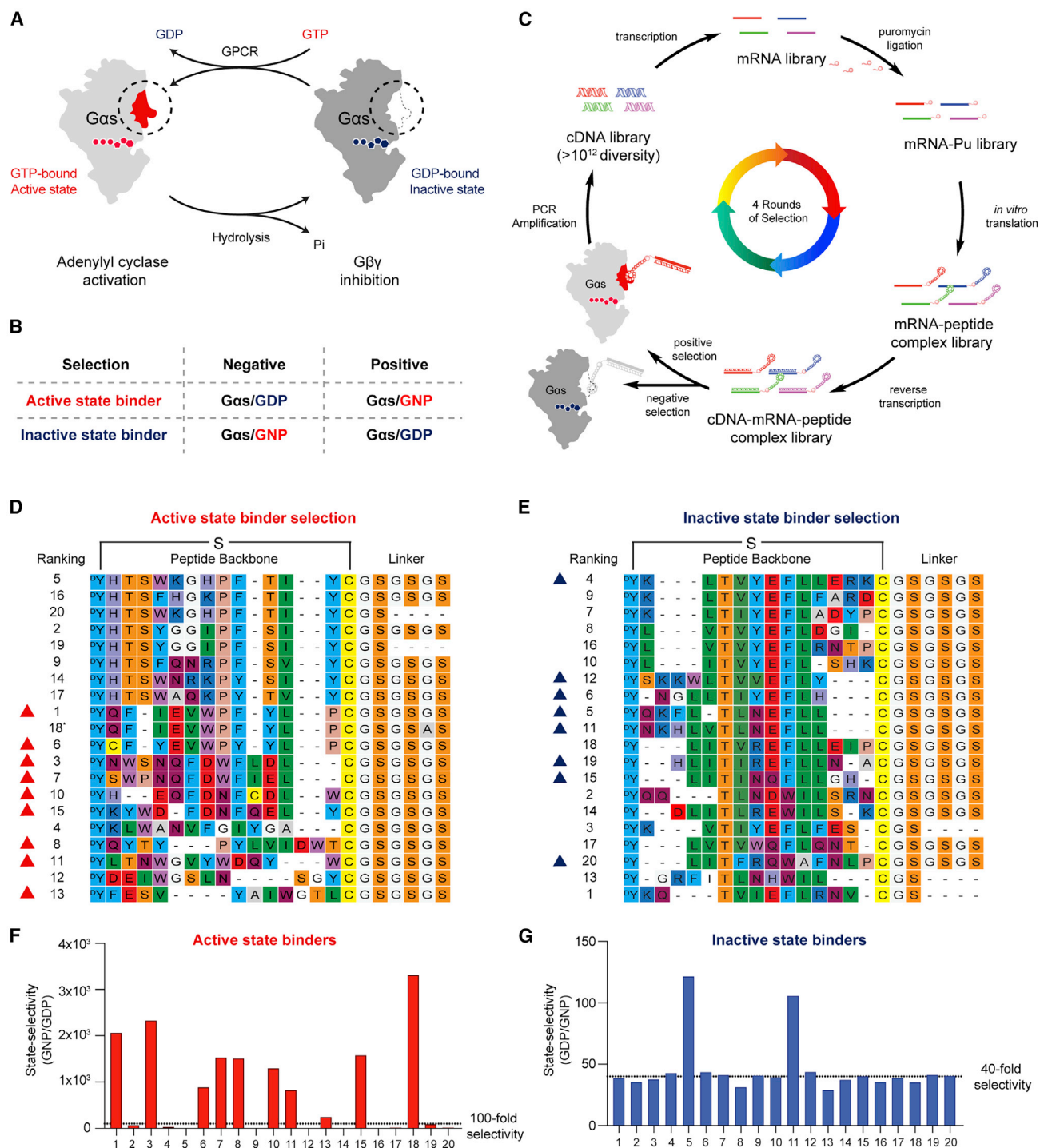


Figure 1. RaPID selection of state-selective Gαs binding cyclic peptides

(A) Gαs adopts distinct conformations, governed by its nucleotide binding state. Switch regions are highlighted with a circle.

(B) A selection strategy to achieve state-selectivity of Gαs binders.

(C) Illustration of the RaPID selection. e.g., Gαs active-state binder selection, positive selection, Gαs/GNP (light gray); negative selection, Gαs/GDP (dark gray).

(D and E) Sequence alignment of top 20 cyclic peptides from the R4 pools. The 18th hit (D, asterisk) was not selected because it has the same core sequence as the first peptide.

(F and G) State selectivity was determined by comparing peptide-mRNA-cDNA complex binding to GDP- or GNP-bound Gαs. Cyclic peptides with high selectivity are marked with triangles and were selected for solid phase synthesis.

See also Figure S1.

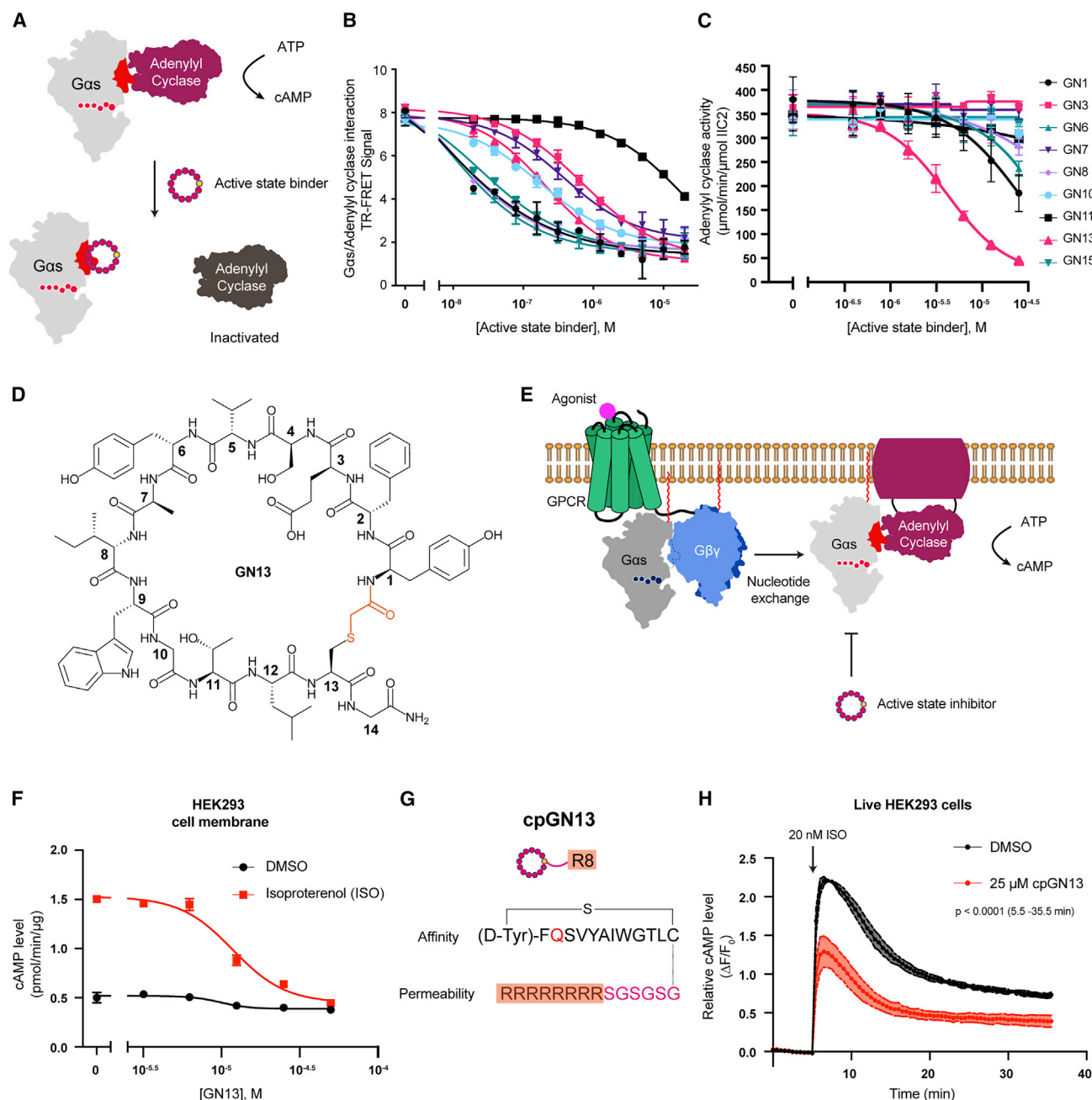


Figure 2. $G_{\alpha s}$ active-state inhibitor GN13 inhibits $G_{\alpha s}$ -mediated adenylyl cyclase activation

(A) Illustration of active-state binders inhibiting $G_{\alpha s}$ -mediated AC activation.

(B) Active-state binders inhibited PPI between $G_{\alpha s}$ /GNP and AC. Mean $\pm$ SD, $n = 3$.

(C) $G_{\alpha s}$ /GNP-mediated AC activation was inhibited by active-state binders. Mean $\pm$ SE, $n = 3$.

(D) Structure of the resynthesized cyclic peptide GN13.

(E) Illustration of GN13 inhibiting GPCR-stimulated $G_{\alpha s}$ /AC activity in cells.

(F) GN13 inhibited ISO-stimulated cAMP production in HEK293 cell membranes. Mean $\pm$ SD, $n = 3$.

(G) Design of a cell-permeable GN13 analog, cpGN13.

(H) Pretreatment with cpGN13 for 10 min inhibited ISO-stimulated cAMP production in live HEK293 cells. Mean $\pm$ SD, $n = 3$.

Two-tailed unpaired t tests (data after 5 min).

See also Figure S2.

chloroalkane-tagged dye (ct-dye), and analyzed by flow cytometry. A lower ct-dye fluorescent signal indicates competition from a higher cytosolic concentration of ct-molecule. We conjugated a chloroalkane tag at the carboxyl terminus (C-term) of GN13 to make ct-GN13-E3Q (Figure S2G). ct-GN13-E3Q exhibited similar biochemical activity to unmodified GN13 (Figure S2H) and showed measurable cell penetration and mild *G α s* inhibition in live cells (Figures S2I and S2J). These results indicated that E3Q mutation and C-term modification of GN13 improved its cell permeability while maintaining *G α s* interaction. We further augmented the cellular concentration of GN13-E3Q by adding a polyarginine motif (R8), a widely used cell-penetrating peptide, at the C-term of GN13 (Figure 2G) (Bechara and Sagan, 2013). Cell-permeable (cp)GN13 significantly inhibited ISO-mediated cAMP production in live HEK293 cells (Figure 2H). Our results demonstrated that GN13 and its cp analogs can modulate β 2AR agonist-stimulated *G α s* activity.

The crystal structure of GppNHp-bound *G α s* in complex with GN13

GN13 inhibited both short and long isoforms of *G α s* in the AC activity assay (Figure S3A). To elucidate how GN13 binds to *G α s* and inhibits *G α s*-mediated AC activation, we solved a co-crystal structure of the GppNHp-bound short isoform *G α s*/GN13 complex. The structure was determined by molecular replacement and refined to 1.57 Å (Figure 3A, see also Table S1). GN13 assumes a highly ordered structure through extensive hydrogen bonding networks with three well-defined water molecules (Figures S3B–S3D). One molecule of GN13 binds to the switch II/ α 3 helix pocket in *G α s* through hydrogen bonding and hydrophobic interactions (Figures 3B and 3C). Specifically, the side chain of GN13 accepts a hydrogen bond (H-bond) from *G α s* K274; the indole ring of GN13 W9 donates an H-bond to *G α s* E268; and the main chains of V5, W9, and T11 in GN13 form H-bonds with *G α s* N279, R280, R231, R232, and S275 (Figure 3B). The side chains of GN13 I8 and W9 (IW motif) dock into two *G α s* hydrophobic pockets (Figure 3C). To validate these hypothesized interactions, we generated *G α s* and GN13 mutants and measured their binding. Although GN13 E3Q mutant retained activity (Figures S2G and S2H), disruption of the H-bond between GN13 $\times 10^3$ and *G α s* K274 with alanine mutations reduced binding by 50% (Figures S3F and S3G). The requirement of a precise *G α s*/GN13 H-bond network was confirmed by a series of *G α s* mutants (R231A, R232A, E268A, K274A, and N279A) (Figure S3G). Finally, the I8A and W9A mutants of GN13 completely abolished GN13 binding, underscoring the importance of the hydrophobic IW motif (Figure S3F).

Residue W9 in GN13 is centrally located at the interface between GN13 and *G α s* (Figures 3B and 3C). In analogous interactions, F991 in AC II (effector of *G α s*, Tesmer et al., 1997), W70 in PDE γ (effector of *G α t*, Slep et al., 2001), and F108 in Nb35 (*G α s*-binding nanobody, Rasmussen et al., 2011) contact the same switch II/ α 3 clefts of *G α s* and *G α t* (Figure 3E, see also Figures S3H and S3I). Comparison between the *G α s*/GN13 structure and the *G α s*/AC complex structure (PDB: 1AZS) suggests that GN13 directly occludes the *G α s*/AC interaction, which accounts for the inhibitory effect of GN13 (Figure 3E).

Structural basis for the nucleotide-state selectivity of GN13

The *G α s*/GNP/GN13 structure strongly resembles the *G α s*/GTP γ S structure (Sunahara et al., 1997), suggesting that GN13 recognizes the active conformation and does not induce significant conformational change upon binding (Figure 3D). GN13 also inhibited oncogenic *G α s* mutants (R201C, R201H, R201S, and Q227L) (Figure S3J), which are locked in the active state by catalytic-site mutations (Hu and Shokat, 2018). However, our structure is much less similar to the structure of inactive *G α s*/GDP (chain I in PDB: 6EG8, Liu et al., 2019). The N terminus of switch II in *G α s*/GDP is unstructured and adjacent to the α 3 helix, with nearly half of the GN13/*G α s* interface disrupted (Figure S3K). In particular, R232 of switch II in *G α s*/GDP is predicted to create a steric clash with GN13 I8, explaining the state selectivity of GN13 for the active state of *G α s*.

To assess the cellular specificity of GN13, we designed a GN13-resistant *G α s* mutant. We examined the structures of *G α s*/GN13 and *G α s*/AC and noted that *G α s* S275 closely contacts with GN13, but not with AC (Figures 3F and 3G). Mutating S275 to a bulkier residue may create a *G α s* mutant that blocks interaction with GN13 but has little effect on AC activation. Indeed, the *G α s* S275L mutant maintained a comparable biochemical activity but was resistant to GN13 (Figure 3H, see also Figure S3L). We tested GN13 in the membranes of GNAS-knockout (GNAS-KO) HEK293 cells that did not express endogenous *G α s* protein (Stallaert et al., 2017). GN13 inhibited ISO-mediated cAMP production in GNAS KO cell membranes transiently expressing reintroduced WT *G α s*, but the inhibitory effect of GN13 was abolished with the *G α s* S275L mutation (Figure 3I). These data demonstrate that the observed cellular activity is due to GN13 binding to the switch II/ α 3 pocket in *G α s*.

Inactive-state binding cyclic peptide GD20 is a *G α s* specific guanine nucleotide dissociation inhibitor

G α GTPase activity hydrolyzes GTP to GDP and rearranges the switch regions to adopt an inactive conformation. This conformation prevents GDP release, which makes GDP dissociation the rate-limiting step of G protein activation (Dror et al., 2015) (Figure 4A, left). To understand how inactive-state binders control *G α s* function, we evaluated the steady-state GTPase activity of *G α s* in the presence of inactive-state binders (Figure 4B). Re-synthesized inactive-state binders are indicated with a “GD” (*G α s*/GDP) preceding their ranking number. All of the tested GD peptides strongly inhibited *G α s* steady-state GTPase activity. GD20 showed the greatest inhibition, with an IC₅₀ of 1.15 $\pm$ 0.16 μ M (Figures 4B and 4C, see also Figure S4A). GD20 also inhibited the long isoform of *G α s*, with an IC₅₀ of 1.32 $\pm$ 0.17 μ M (Figure S4F).

Interestingly, GN13 modestly increased *G α s* steady-state GTPase activity (Figure S4A). To understand how GD20 and GN13 regulate *G α s* enzymatic activity, we determined rate constants for both GDP dissociation and GTP γ S binding. GD20 drastically reduced the GDP dissociation rates (k_{off}) and the apparent rate of GTP γ S binding (k_{app}), indicating that GD20 is a guanine nucleotide dissociation inhibitor (GDI) (Figures 4D and 4E). On the contrary, GN13 only slightly influenced *G α s* GDP dissociation (Figure S4B), and instead slightly increased

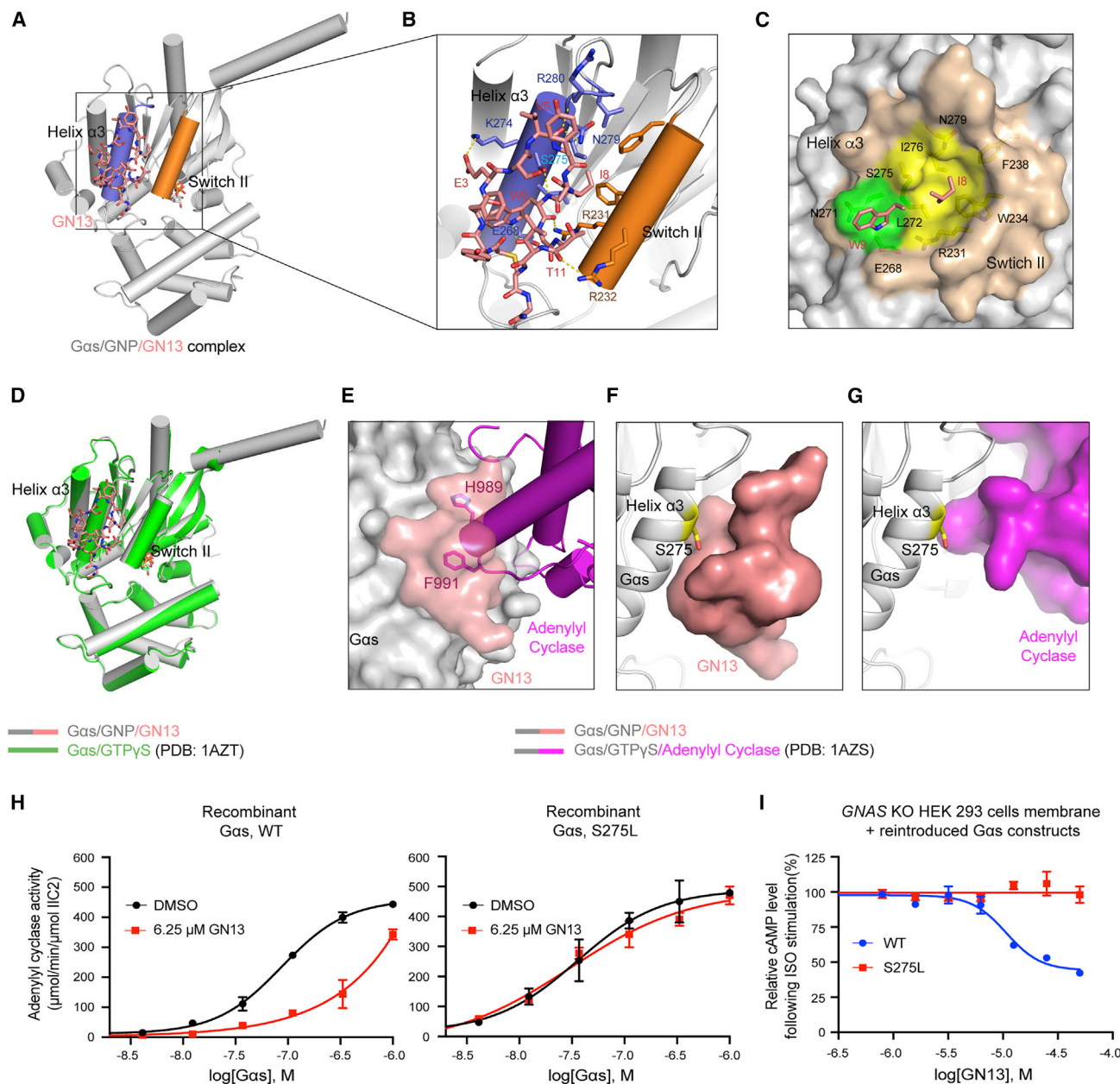


Figure 3. The crystal Structure of GppNHP-bound $G\alpha_s$ in complex with GN13

(A) Overall structure of the $G\alpha_s$ /GNP/GN13 complex. GN13 (salmon) binds in between switch II (orange) and the $\alpha 3$ helix (slate).
 (B) Structural details of $G\alpha_s$ /GN13 interaction. H-bonds are shown as yellow dashed lines.
 (C) Close-up view of two $G\alpha_s$ hydrophobic pockets (green and yellow) that accommodate I8 and W9 of GN13 (salmon). $G\alpha_s$ residues that form those pockets are shown as sticks.
 (D) Alignment of $G\alpha_s$ /GN13 structure (gray) with the structure of $G\alpha_s$ /GTP γ S (green, PDB: 1AZT). Root-mean-square deviation = 0.479 Å.
 (E) Our $G\alpha_s$ /GN13 (gray/salmon) structure was superimposed on the $G\alpha_s$ /AC complex structure (gray/magenta, PDB: 1AZS). GN13 blocks H989/F991 of AC from binding to the same pocket in $G\alpha_s$.
 (F and G) Close-up view of the interaction between GN13 (salmon) and the $G\alpha_s$ $\alpha 3$ helix (F) and the interaction between AC (magenta) and the $G\alpha_s$ $\alpha 3$ helix (G, PDB: 1AZS). S275 is shown as sticks.
 (H) $G\alpha_s$ WT and $G\alpha_s$ S275L have comparable biochemical activities in the AC activation assay (black). GN13 inhibited AC activation by $G\alpha_s$ WT (red, left) but not by $G\alpha_s$ S275L (red, right). Mean $\pm$ SD, $n = 3$.
 (I) $G\alpha_s$ S275L confers resistance to GN13 inhibition in HEK293 cell membranes. Mean $\pm$ SD, $n = 3$.
 See also [Figure S3](#) and [Table S1](#).

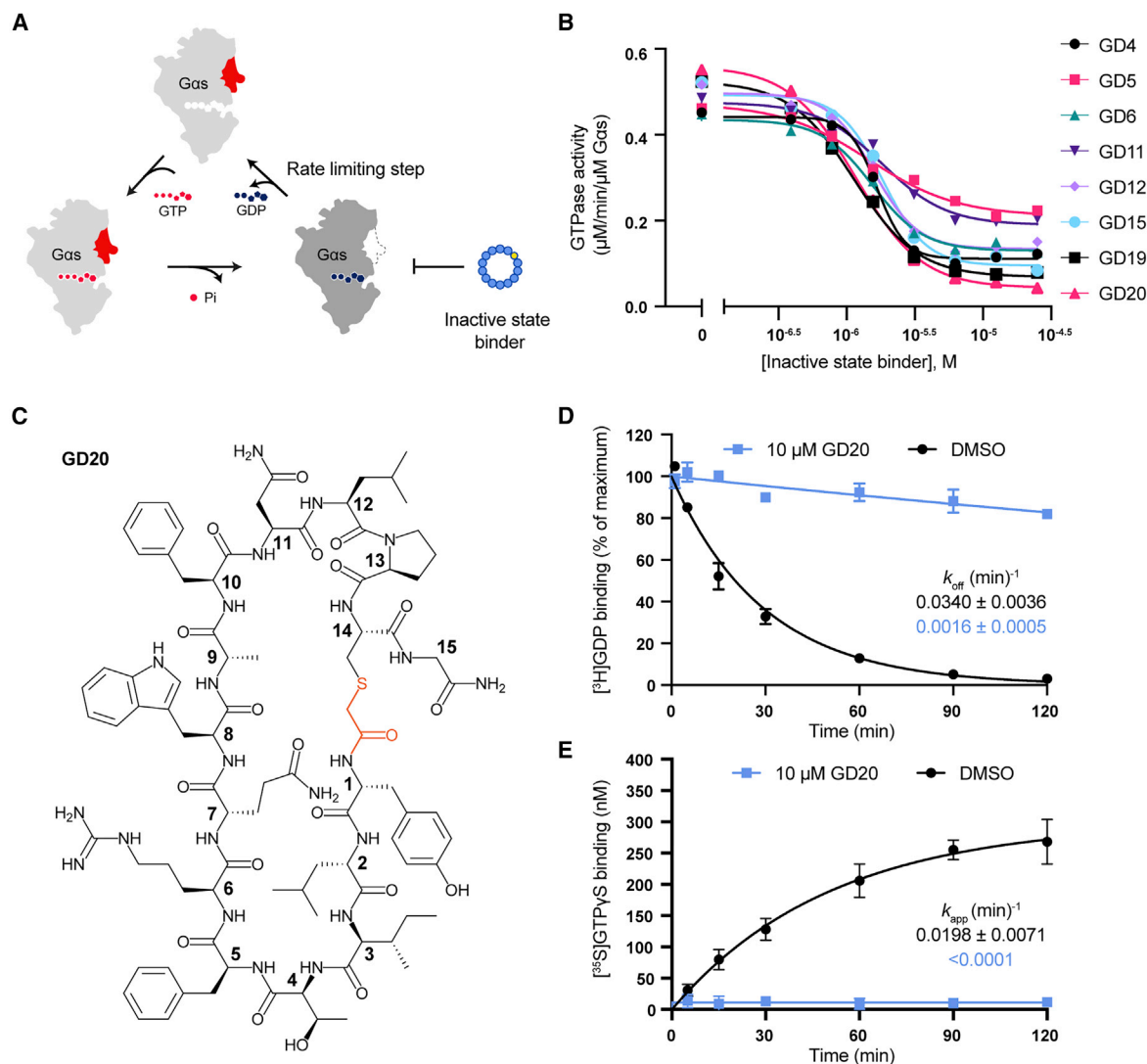


Figure 4. Inactive-state binding cyclic peptide GD20 is a $G\alpha_s$ specific guanine nucleotide dissociation inhibitor

(A) Illustration of inactive-state binders inhibiting $G\alpha_s$ steady-state GTPase activity.

(B) $G\alpha_s$ steady-state GTPase activity was inhibited by GD peptides. The data represent one measurement. $G\alpha_s$ steady-state GTPase activity in the presence of GD20 was repeated twice in Figure S4A.

(C) Structure of the resynthesized cyclic peptide GD20.

(D) GD20 inhibited $G\alpha_s$ GDP dissociation. Mean $\pm$ SD, $n = 3$.

(E) GD20 inhibited GTP γ S binding to $G\alpha_s$. Mean $\pm$ SD, $n = 3$.

See also Figure S4.

the maximum GTP γ S binding (Figure S4C). The discrepancy between GD20 and GN13 exemplifies how state-selective $G\alpha_s$ binders fine-tune $G\alpha_s$ enzymatic activity. This precise regulation also appears at the G protein family level. $G\alpha_i$ was much less sensitive to GD20 and GN13 (Figures S4D and S4E), highlighting the class-specificity of both cyclic peptides.

The crystal structure of GDP-bound $G\alpha_s$ in complex with GD20

To explore how GD20 favors $G\alpha_s$ /GDP and inhibits GDP dissociation, we solved a structure of the $G\alpha_s$ /GDP/GD20 complex.

The structure was determined by molecular replacement and refined to 1.95 Å (Figure 5A, see also Table S2). Four well-defined water molecules and a number of intramolecular H-bonds constructed a helical secondary structure in GD20 (Figure S5A–S5E). One molecule of GD20 binds to the switch II/ $\alpha 3$ pocket in $G\alpha_s$ through electrostatic interactions, H-bonds, and hydrophobic interactions (Figures 5B and 5C). Specifically, the side chain of GD20 R6 forms a salt bridge with $G\alpha_s$ E268, and this ion pair is stabilized by $G\alpha_s$ N271; the main chain carbonyl oxygen of GD20 A9 forms an H-bond network with $G\alpha_s$ S275 and N279; and the main chain of D229 and the side chain of R231

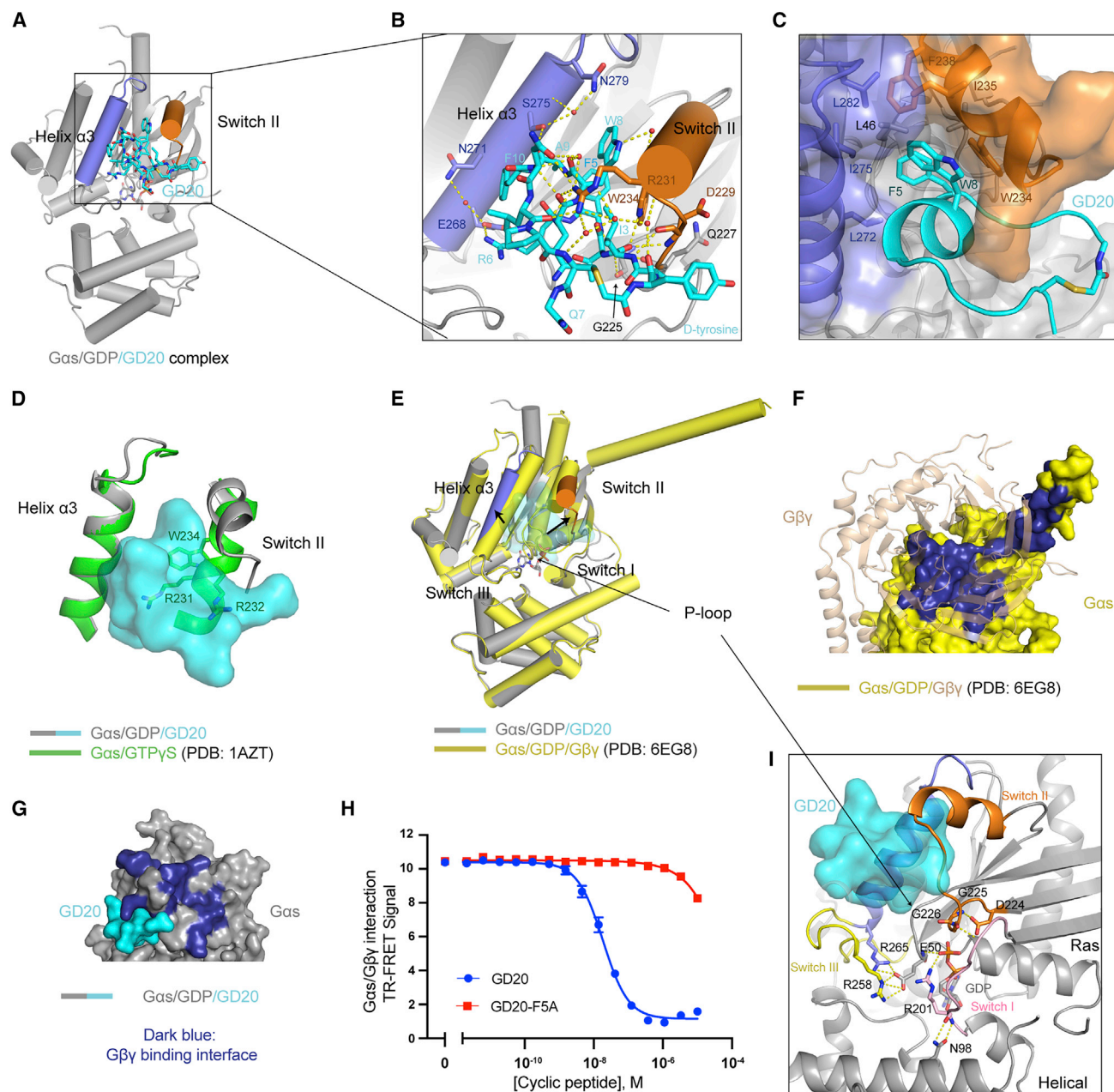
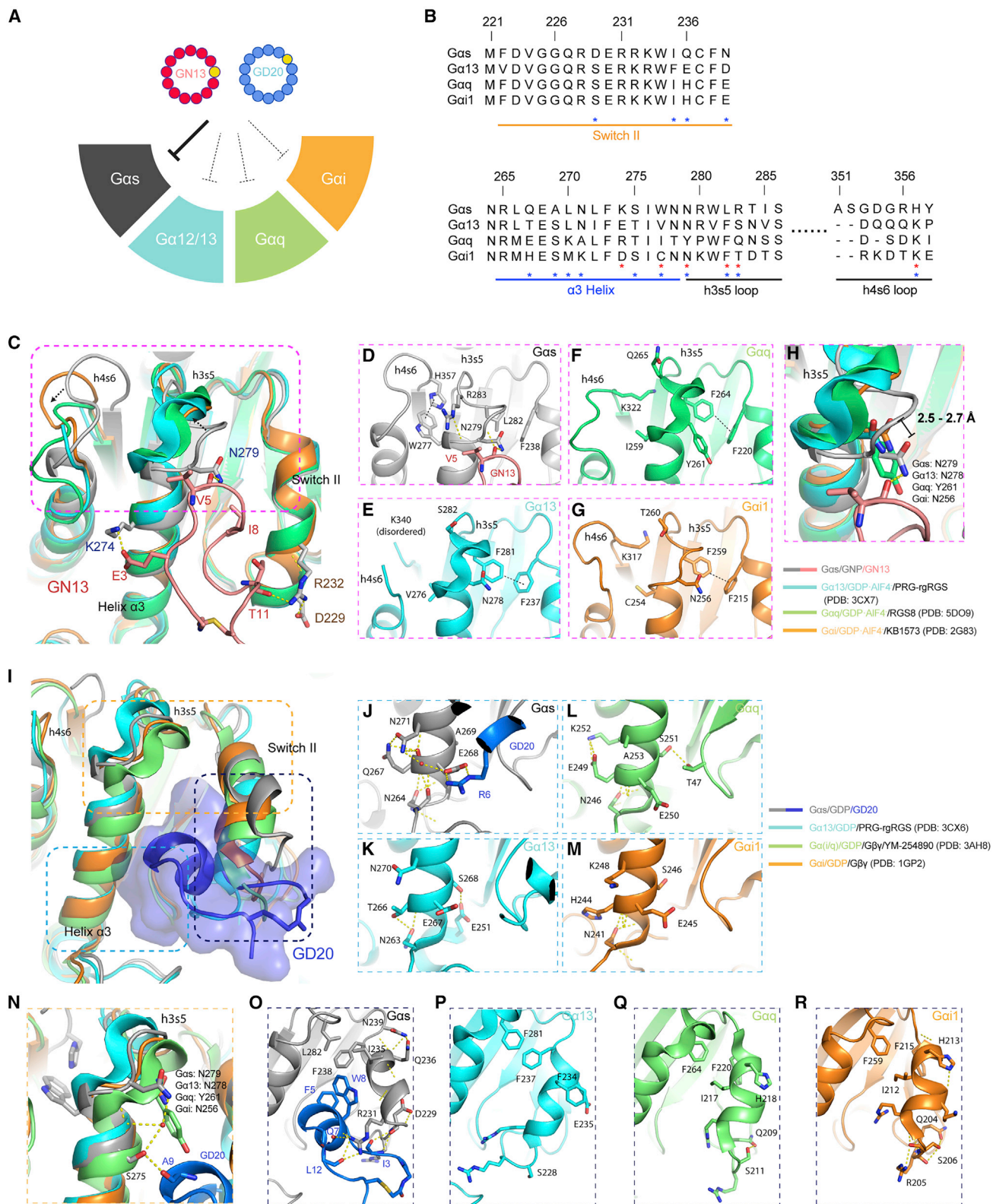


Figure 5. The crystal structure of GDP-bound $G\alpha s$ in complex with GD20

(A) Overall structure of the $G\alpha s$ /GDP/GD20 complex. GD20 (cyan) binds in between switch II (orange) and the $\alpha 3$ helix (slate).
 (B) Structural details of $G\alpha s$ /GD20 interaction. Ion pair and H-bonds are shown as yellow dashed lines.
 (C) Close-up view of a hydrophobic pocket in $G\alpha s$ that accommodates GD20 F5 and W8 (cyan). $G\alpha s$ residues that form the hydrophobic pocket are shown as sticks.
 (D) Alignment of $G\alpha s$ /GD20 complex structure (gray) with the structure of $G\alpha s$ /GTP γ S (green, PDB: 1AZT) in the switch II/ $\alpha 3$ pocket.
 (E) Alignment of $G\alpha s$ /GD20 complex structure (gray) with the structure of $G\alpha s$ /GDP (yellow) in the structure of $G\alpha s$ /G $\beta 1$ / $\gamma 2$ heterotrimer (PDB: 6EG8). G $\beta \gamma$ was hidden for clarity.
 (F) Structural details of the $G\alpha s$ (yellow, surface) and G $\beta \gamma$ (wheat, cartoon) binding interface (dark blue) (PDB: 6EG8).
 (G) The G $\beta \gamma$ binding interface (dark blue) of $G\alpha s$ is rearranged when GD20 (cyan) binds to $G\alpha s$ (gray). G $\beta \gamma$ was hidden for clarity.
 (H) GD20, but not GD20-F5A, inhibited PPI between $G\alpha s$ /GDP and G $\beta \gamma$ (C68S). Mean $\pm$ SD, $n = 3$.
 (I) Close-up view of $G\alpha s$ nucleotide binding pocket in our $G\alpha s$ /GD20 complex structure. Residues that stabilize GDP binding are shown as sticks.
 See also [Figure S5](#), [Table S2](#), and [Video S1](#).



(legend on next page)

and W234 in *Gαs* coordinate a complex H-bond network with I3, W8, N11, L12, C14, and D-tyrosine in GD20 (Figure 5B). These interactions rearrange the flexible *Gαs* switch II and bury GD20 F5 and W8 inside of a hydrophobic pocket (Figure 5C).

GD20 binds to *Gαs*/GDP with a K_D value of 31.4 ± 0.7 nM (Figure S5F, see also Table S3). Single point mutations of GD20, including F5A, R6A, and W8A, nearly completely abolished *Gαs* binding, confirming the importance of these residues (Figure S5G). The hypothesized interactions were further validated by *Gαs* mutagenesis studies (Figure S5H): *Gαs* mutations at contact residues (D229A, R231A, E268A, N271A, and N279A) eliminated GD20 binding to different extents, while mutations at non-contact residues (K274A and R280A) did not influence GD20 binding. The R232A mutation may indirectly reduce GD20 binding through perturbing the switch II conformation.

Structural basis for the nucleotide-state selectivity and biochemical activity of GD20

GD20 showed high nucleotide-state selectivity for the GDP-bound *Gαs* (Figure S5I). To understand the mechanism for this selectivity, we superimposed our *Gαs*/GD20 structure on the structure of active GTPγS-bound *Gαs* (Sunahara et al., 1997). The rigidified switch II in *Gαs*/GTPγS—R231, R232, and W234 in particular—clashes with GD20 (Figure 5D). Indeed, GD20 did not inhibit active-state *Gαs*-mediated AC activation in biochemical or cell membrane experiments (Figures S5J and S5K). Next, we compared our *Gαs*/GD20 structure with a structure of *Gαs*/GDP in complex with Gβγ (chain I in PDB: 6EG8, Liu et al., 2019) (Figure 5E). The structural motifs in *Gαs* (such as switch I, III, and the P loop) that are critical for GDP binding remain unchanged, highlighting the GDP-state selective nature of GD20. However, GD20 binding induces a significant conformational shift at the Gβγ-binding surface by expanding the switch II/α3 pocket. Hence, GD20 may block Gβγ binding to *Gαs* in a potentially competitive manner (Figures 5F and 5G, see also Video S1). We measured the interaction between *Gαs*/GDP and Gβγ in the presence of GD20 or a *Gαs* binding deficient analog, GD20-F5A, using a FRET assay (Figure S5L). Indeed, GD20, but not GD20-F5A, showed potent inhibition of the *Gαs*/Gβγ interaction, with an IC_{50} of 18.4 ± 2.0 nM (Figure 5H).

The *Gαs*/GD20 structure also illuminates GD20 GDI activity (Figure 5I). GDP dissociation from *Gαs* requires conformational

changes that weaken GDP affinity and Ras/Helical domain separation to allow GDP release (Dror et al., 2015). GD20 does not engage the GDP exit tunnel and so does not directly occlude GDP release. Instead, GD20 phenocopies the effects of Gβγ GDI activity, stabilizing the conformations of switch I, III, and the P loop in the GDP-bound state. Such a conformational lock not only orients *Gαs* R201 and E50 to directly capture the β-phosphate of GDP, but also inhibits the spontaneously Ras/Helical domain separation by stabilizing H-bonds between *Gαs* R201 and N98. As a result, GD20 antagonizes GDP dissociation from *Gαs*.

G protein class-specificity of GN13 and GD20

There are four main families of *Gα* proteins: *Gαs*, *Gαi*, *Gα12/13*, and *Gαq*. These *Gα* proteins are structurally similar, yet they transduce divergent G protein-coupled receptor (GPCR) signaling activation by binding to distinct effectors (Syrovatkina et al., 2016). To assess whether GN13 and GD20 can distinguish *Gαs* from other *Gα* proteins, we measured their binding to *Gαi*, *Gα13*, and *Gαq*. In contrast to their strong binding to *Gαs* (Figure S2C and S5F), GN13 and GD20 showed little to no detectable binding to either nucleotide state of *Gα13*, *Gαq*, and *Gαi* at the highest concentration tested (Figure S6A–S6G, and S6J). Furthermore, GD20 disrupted *Gα*/Gβγ interaction at least 100-fold selectively for *Gαs* over *Gαi* (Figure S6K). These results demonstrate that GN13 and GD20 possess excellent G protein class specificity.

To identify G protein specificity determinants of both cyclic peptides, we aligned *Gα* sequences at the cyclic peptide binding interfaces (Figure 6B) and compared our structures with structures of other *Gα* proteins in complex with effectors or peptide inhibitors (Chen et al., 2008; Taylor et al., 2016; Johnston et al., 2006; Nishimura et al., 2010; Wall et al., 1995) (Figures 6C–6R).

GDP- AlF_4^- -bound active structures of *Gα13*, *Gαq*, and *Gαi* were superimposed on *Gαs*/GNP in our *Gαs*/GN13 complex (Figure 6C). There were a few profound differences among *Gα* proteins. (1) A distinctive π-π stacking between W277 and H357 and a unique R283 in *Gαs* (the WHR triad) define the positions of the h3s5 and h4s6 loops, which present N279 to form an H-bond with GN13 V5 (Figures 6D and 6H). Changes of the triad in *Gα13* (VKS), *Gαq* (IKQ), and *Gαi* (CKT) may alter the conformation of loop h3s5 to disrupt interaction with GN13

Figure 6. G protein class-specificity of GN13 and GD20

- (A) GN13 and GD20 are class-specific *Gαs* inhibitors.
(B) Sequence alignment of *Gα* proteins around the cyclic peptide binding site. The residue numbering is based on *Gαs*. Residues that determine the specificity of GN13 (red) or GD20 (blue) are marked with asterisks.
(C) The active states of *Gα13* (cyan, PDB: 3CX7), *Gαq* (green, PDB: 5DO9), and *Gαi* (orange, PDB: 2G83) from their complex structures were superimposed on *Gαs*/GNP in our *Gαs* (gray)/GN13 (salmon) structure.
(D–G) Structural details of the GN13 binding pocket in different active state *Gα* proteins. H-bonds are shown as yellow dashed lines. CH/π interactions are shown as black dashed lines.
(H) Close-up view of the critical N279 in *Gαs*/GNP. Homologous residues in other *Gα* proteins are labeled with different colors. The distances between the Cα of *Gαs* N279 and the Cα of other homologous residues are indicated.
(I) The inactive states of *Gα13* (cyan, PDB: 3CX6), *Gαq* (green, PDB: 3AH8), and *Gαi* (orange, PDB: 1GP2) from their complex structures were superimposed on *Gαs*/GDP in our *Gαs* (dark gray)/GD20 (blue) structure.
(J–M) Structural details of the α3 helices in different *Gα*/GDP proteins.
(N) Close-up view of the critical N279 in *Gαs*/GDP. Homologous residues in other *Gα* proteins are labeled with different colors.
(O–R) Structural details of the switch II regions in different *Gα*/GDP proteins.
See also Figure S6.

(Figures 6E–6G). The replacement of N279 by Y261 in $G\alpha_q$ may further limit the $G\alpha_q$ /GN13 interaction (Figure 6H). (2) GN13 I8 docks between F238 and L282 in $G\alpha_s$. Substitution of L282 with a phenylalanine residue in $G\alpha_{13}$, $G\alpha_q$, and $G\alpha_i$ sterically reshapes this hydrophobic pocket (Figures 6E–6G). The same hydrophobic pocket also controls the binding of other $G\alpha$ effectors (Chen et al., 2005; Tesmer et al., 2005). (3) $G\alpha_s$ K274 interacts with the negatively charged GN13. Homologous residues in $G\alpha_{13}$ (E273) and $G\alpha_i$ (D251) repel the negative charge of GN13. (4) The unique $G\alpha_s$ D229 might participate in an H-bond interaction between GN13 T11 and $G\alpha_s$ R232. However, $G\alpha_s$ D229A mutation did not influence GN13 binding (Figure S3G).

To explain GD20 $G\alpha$ selectivity, we compared our $G\alpha_s$ /GD20 structure with GDP-bound $G\alpha_{13}$, $G\alpha_q$, and $G\alpha_i$ (Figure 6I). The specificity of GD20 is determined by three major contacts which involve electrostatic interactions, van der Waals interactions, and hydrogen bonding. (1) $G\alpha_s$ N271 positions E268 to interact with GD20 R6 (Figure 6J). Rewired α 3 helix H-bond networks in other $G\alpha$ proteins disfavor this salt bridge formation (Figures 6K–6M). (2) The WHR triad orients N279 in $G\alpha_s$ loop h3s5 for better GD20 binding. N279 and S275 form water-mediated H-bonds with A9 in GD20 (Figure 6N). (3) Switch II has several notable differences between $G\alpha$ proteins. Three unique residues (D229, Q236, and N239) and a conserved R231 in $G\alpha_s$ support a helical structure in switch II and form H-bond networks with GD20 I3, Q7 and L12 (Figure 6O). Supporting this model, the $G\alpha_s$ D229A and R231A mutants reduced GD20 binding (Figure S5H). The dynamic $G\alpha_s$ switch II also shapes a distinctive hydrophobic pocket (L282–F238–I235) for engagement of GD20. Homologous residues in $G\alpha_{13}$, $G\alpha_q$, and $G\alpha_i$ adopt conformations not compatible with GD20 binding (Figure 6P–6R). In summary, sequence alignment and structural analysis revealed that some of the $G\alpha_s$ residues that directly interact with GN13 and GD20 are not conserved in other $G\alpha$ proteins, which explains the G protein class specificity of both cyclic peptides.

A cell-permeable GD20 analog, cpGD20, is a dual-effect G protein modulator

GPCR signaling releases GTP-bound $G\alpha$ and free $G\beta\gamma$ to engage their own effectors for downstream signaling. $G\alpha$ /GDP is a functional “OFF” switch: it tightly reassociates with $G\beta\gamma$, masking effector-binding surfaces on both $G\alpha_s$ and $G\beta\gamma$ (Gulati et al., 2018). We hypothesized that GD20, a $G\alpha_s$ / $G\beta\gamma$ protein-protein interaction (PPI) inhibitor, may both block $G\alpha_s$ / $G\beta\gamma$ reassociation following receptor stimulation and prolong $G\beta\gamma$ -mediated effector activation (Figure 7A).

We first tested the cell permeability of GD20. The C-terminal residue of GD20 was conjugated with a chloroalkane tag to make ct-GD20 (Figure S7A). Although ct-GD20 is cell permeable, an F10L substitution further improved cell penetration (Figure 7B, see also Figures S7B and S7C). cpGD20 (GD20-F10L) retains binding affinity for $G\alpha_s$ /GDP with a K_D value of 14.5 ± 0.4 nM (Figure S7D, see also Table S3), and showed biochemical activity, state selectivity, and class specificity comparable to GD20 (Figure S7E, and S6H–S6J). cpGD20 disrupted $G\alpha_s$ / $G\beta\gamma$ interaction with an IC_{50} of 14.0 ± 0.6 nM and exhibited almost 100-fold selectivity over $G\alpha_i$ (Figure S6L).

cpGD20 did not inhibit $G\alpha_s$ -mediated cAMP production in live HEK293 cells, confirming its nucleotide-state-selectivity (Figure S7F). We tested whether cpGD20 could inhibit $G\alpha_s$ / $G\beta\gamma$ interaction in HEK293 cells using a bioluminescence resonance energy transfer (BRET2) assay (Olsen et al., 2020). Rluc8 was inserted into a flexible loop region between the α B- α C helices of $G\alpha_s$ ($G\alpha_s$ -Rluc8) and GFP2 was inserted at the N terminus of $G\gamma_2$ (GFP2- $G\gamma$) to capture $G\alpha\beta\gamma$ heterotrimer interaction. A decrease in BRET signal indicates $G\alpha\beta\gamma$ dissociation (Figures S7G and S7H). In cells transiently transfected with β_2 AR, $G\alpha_s$ -Rluc8, $G\beta_1$, and GFP2- $G\gamma_2$, ISO stimulated a reference net BRET response. Pretreatment with cpGD20 induced greater net BRET signal between $G\alpha_s$ and $G\beta\gamma$ (Figure 7C, see also Figure S7I). In comparison, the $G\alpha_s$ binding-deficient cpGD20-F5A failed to induce a larger BRET response (Figure 7C). To assess the specificity of cpGD20 at the G protein level, we tested it against $G\alpha_i$ / $G\beta\gamma$. HEK293 cells transiently expressing $G\alpha_i$ -coupled muscarinic acetylcholine receptor M2 (M2R), $G\alpha_i$ -Rluc8, $G\beta_1$, and GFP2- $G\gamma_2$ were stimulated with the M2R agonist, acetylcholine (ACh). Pretreatment with cpGD20 did not induce a net BRET signal change (Figure 7D). These data suggest that cpGD20 can specifically capture monomeric $G\alpha_s$ after G protein activation and block $G\alpha_s$ / $G\beta\gamma$ reassociation.

We investigated whether cpGD20 could prolong $G\beta\gamma$ -mediated effector activation after $G\alpha_s$ / $G\beta\gamma$ dissociation. We focused on a well-studied $G\beta\gamma$ effector: The G protein-activated inward rectifier K^+ (GIRK) channel, which produces inward K^+ current upon $G\beta\gamma$ binding. Voltage-clamp experiments of HEK293 cells transiently transfected with β_2 AR, overexpressed $G\alpha_s$ / $G\beta\gamma$ trimer, and GIRK4 showed GIRK activation upon ISO stimulation (Figure 7E, black), consistent with previous findings (Touhara and MacKinnon, 2018). Cyclic peptides treatment did not attenuate the amplitude of GIRK activation (Figures 7E and 7G). However, cpGD20, but not cpGD20-F5A or DMSO, significantly delayed GIRK channel deactivation after ISO washout (Figures 7E and 7F). These results suggest that the $G\alpha_s$ -specific inactive-state inhibitor cpGD20 modulates G protein signaling in two ways, liberating $G\alpha_s$ -bound $G\beta\gamma$ while sequestering GDP-bound $G\alpha_s$.

DISCUSSION

GPCRs and G proteins comprise the largest human family of signal transducing proteins. Although ~35% of approved drugs target GPCRs, directly targeting the downstream integrator G proteins has the potential for broader efficacy via blocking convergent pathways shared by multiple GPCRs (Bonacci et al., 2006; Gulati et al., 2018). However, there is a striking absence of drug-like chemical matter that specifically targets the $G\alpha$ proteins in cells. Cyclic peptides bridge the chemical space between small molecules and biologics and being capable of recognizing shallow effector binding pockets at PPI interfaces while also maintaining favorable pharmacological properties. This is demonstrated here by the development of $G\alpha_s$ selective cyclic peptide inhibitors GN13 and GD20 and their analogs, which specifically recognize the $G\alpha_s$ switch II/ α 3 pocket. Peptide cyclization and introduction of a non-canonical

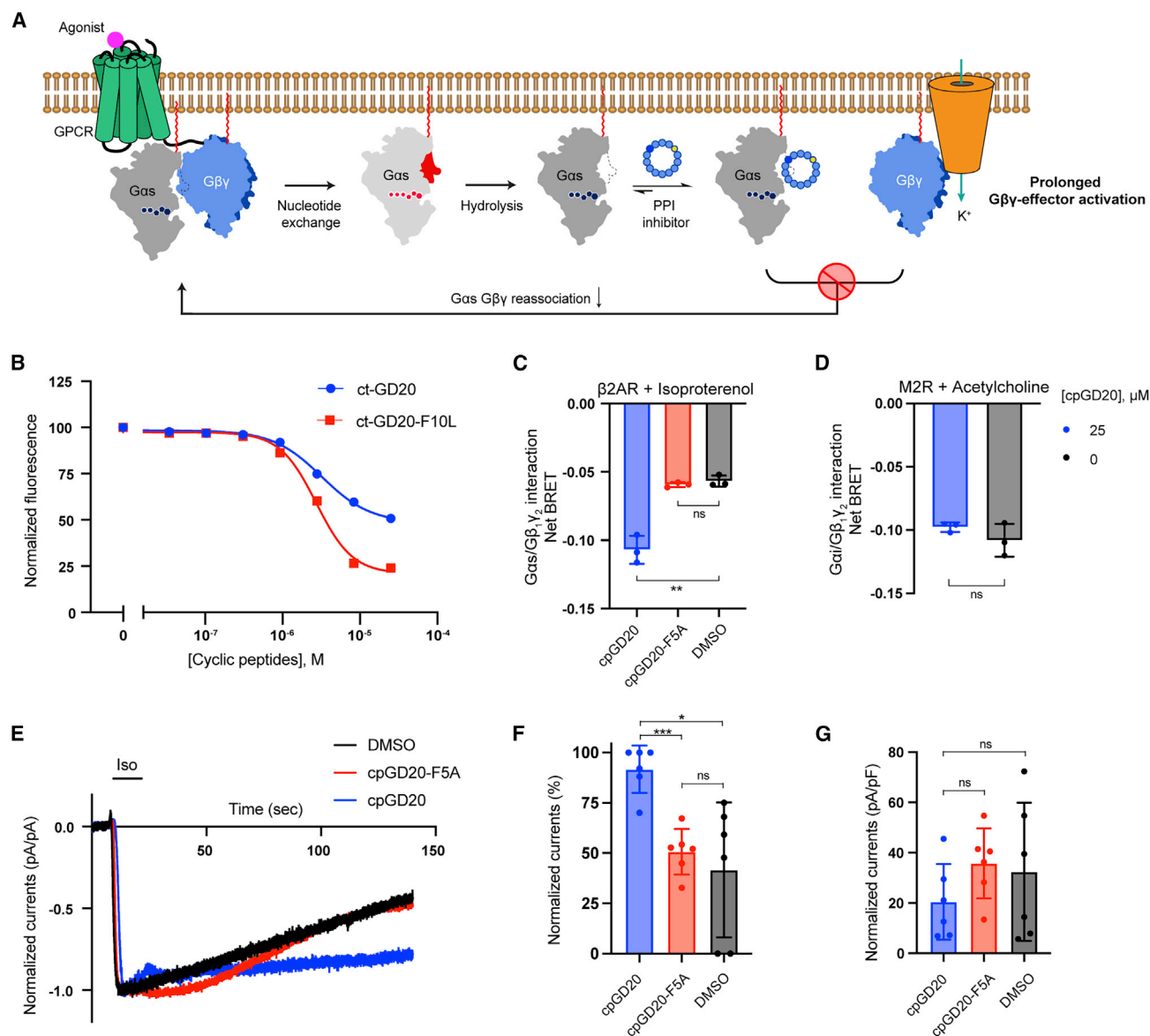


Figure 7. A cell-permeable GD20 analog, cpGD20, is a dual-effect G protein modulator

(A) Illustration of *Gas*/Gβγ PPI inhibitors acting as dual-effect G protein modulators in cells.

(B) CAPA assay results of ct-GD20 and ct-GD20-F10L. Mean ± SD, n = 3.

(C) 25 μM cpGD20, but not 25 μM cpGD20-F5A, inhibited *Gas*/Gβγ reassociation in HEK293 cells transfected with β2AR and *Gas*/Gβ1γ2. *Gas*/Gβγ dissociation was measured by BRET signal reduction after 10 nM ISO application. BRET signal was normalized to cells that were not treated with ISO. Mean ± SD, n = 3. Two-tailed unpaired t tests, **p < 0.01, ns p > 0.05.

(D) cpGD20 did not inhibit *Gαi*/Gβγ reassociation in HEK293 cells transfected with M2R and *Gαi*/Gβ1γ2. *Gαi*/Gβγ dissociation was measured by BRET signal reduction after 100 nM ACh application. BRET signal was normalized to cells that were not treated with ACh. Mean ± SD, n = 3. Two-tailed unpaired t tests, ns p > 0.05.

(E) Representative voltage-clamp recordings of HEK293 cells transiently transfected with β2AR, GIRK4, Gβγ-Venus, and *Gas*. Membrane potential was held at −80 mV. 1 μM ISO was applied as indicated. 25 μM of cpGD20, cpGD20-F5A, or DMSO were added to the pipette solution prior to recordings.

(F) The amounts of residual ISO-activated currents after 60 s of washout normalized to the maximum ISO-activated currents. Mean ± SD, n = 6. Two-tailed unpaired t tests with Welch's correction, *p < 0.05, ***p < 0.001, ns p > 0.05.

(G) Maximum ISO-activated currents normalized to the capacitance of the cells. Mean ± SD, n = 6. Two-tailed unpaired t tests with Welch's correction, *p < 0.05, ns p > 0.05.

See also Figure S7.

amino acid (D-tyrosine) give these $G_{\alpha s}$ inhibitors better cell permeability and chemical stability (Figure 7B, see also Figure S2I, Tables S4 and S5), comparable to small molecule drugs. In contrast to the complex cyclic peptide natural product YM-254890, our $G_{\alpha s}$ -binding cyclic peptides can be easily derivatized through side-chain substitutions. The high-resolution co-crystal structures of $G_{\alpha s}$ with our cyclic peptides enable us to program the protein-inhibitor interaction for desired biological effects. This tunability is exemplified by two GD20 analogs, cpGD20 and GD20-F5A, in that a single point substitution drastically changed the biochemical and pharmacological properties of GD20, providing opportunities for further optimization.

$G_{\alpha s}$ is one of the most frequently mutated G proteins in human cancer. Hotspot mutations in $G_{\alpha s}$ (Q227 and R201) lock $G_{\alpha s}$ in a constitutively active conformation (Zachary et al., 1990; Hu and Shokat, 2018). The cyclic peptide GN13 recognized this particular $G_{\alpha s}$ conformation and inhibited all the tested $G_{\alpha s}$ oncogenic mutants (Q227L, R201C, R201H, and R201S) in the AC activation assay (Figures S3J and S5J). Pharmacologically targeting the $G_{\alpha s}$ active state with GN13 has demonstrated the ligandability of oncogenic $G_{\alpha s}$ and opened up potential to uncover molecular mechanism of tumorigenic $G_{\alpha s}$ signaling.

Both GN13 and GD20 bind at the evolutionally conserved switch II/ $\alpha 3$ pocket. This pocket is generally the site of effector binding, with subtle differences conferred by sequence variability between homologous G_{α} proteins and by binding of different nucleotides. Our diverse chemical library, along with both positive and negative selection, enabled us to survey the sequence space of cyclic peptides and discover selective binders that capture specific conformations of the $G_{\alpha s}$ switch II/ $\alpha 3$ pocket. The resulting $G_{\alpha s}$ -cyclic peptide pairings are highly class-specific and state-selective and thus could be useful for developing biosensors that directly detect $G_{\alpha s}$ /GTP or $G_{\alpha s}$ /GDP in cells (Maziarz et al., 2020). This molecular recognition is complementary to $G_{\alpha s}$ nanobody sensors that only capture nucleotide-free $G_{\alpha s}$ (Manglik et al., 2017).

Pharmacological interrogation of GPCR-mediated signaling events has been largely limited to the receptors. The cell-permeable cyclic peptides cpGN13 and cpGD20 offer an opportunity to directly probe the $G_{\alpha s}$ /G $\beta\gamma$ trimer at the G protein level and represent additional modes of pharmacological intervention in stimulatory GPCR signaling. The active-state inhibitor, cpGN13 inhibits cAMP production in cells by directly competing with the $G_{\alpha s}$ effector, AC. This mechanism is distinct from a commonly used $G_{\alpha s}$ inhibitor, cholera toxin, which catalyzes $G_{\alpha s}$ ADP-ribosylation and leads to transient $G_{\alpha s}$ activation and subsequent degradation (Chang and Bourne, 1989). The inactive-state inhibitor cpGD20 sequesters monomeric $G_{\alpha s}$ /GDP and releases G $\beta\gamma$ from inhibition by $G_{\alpha s}$ /GDP following receptor stimulation. Therefore, cpGD20 could potentially provide a unique approach to elucidate or even rewire $G_{\alpha s}$ -coupled receptor signaling by activating G $\beta\gamma$ -dependent pathways. Moreover, rapid G_{α} /G $\beta\gamma$ reassociation terminates canonical GPCR-dependent G protein signaling within seconds (Ghosh et al., 2017). The slow-dissociating $G_{\alpha s}$ /cpGD20 interaction (Figure S7D and

Table S3) may be able to trap inactive-state $G_{\alpha s}$ and extend G $\beta\gamma$ -dependent signaling.

Our demonstration of the use of the RaPID cyclic peptide platform through both positive and negative selection steps provides proof of principle for a path to discovering cell-permeable, class-specific and state-selective inhibitors of the remainder of the GTPase family.

Limitations of the study

Although GN13, GD20, and their analogs are strong $G_{\alpha s}$ binders, with K_D values in the nanomolar range, their potencies are compromised in cells. This is likely due to the difficulty of competing tight PPIs on cell membranes and the relatively lower cell penetration of cyclic peptides. Optimizing cyclic peptides with non-canonical residues could potentially improve the potency and cp of cpGN13 and cpGD20 to overcome this limitation. Second, we used $G_{\alpha 13}$, $G_{\alpha q}$, and $G_{\alpha i1}$ to test the G protein class specificity of both GN13 and GD20, but we have not performed binding experiments with the entire G_{α} protein family (e.g., $G_{\alpha olf}$, $G_{\alpha 11}$, G12, and others). It will be of interest in the future to test the specificity of GN13 and GD20 against other G_{α} proteins. Last, we investigated the cellular activities of cpGN13 and cpGD20 with two GPCRs, $\beta 2AR$ and M2R, and one G $\beta\gamma$ effector, GIRK4. It would be worthwhile to test more $G_{\alpha s}$ -coupled receptors and G $\beta\gamma$ effectors to further explore the scope of their utility.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- KEY RESOURCES TABLE
- RESOURCE AVAILABILITY
 - Lead contact
 - Materials availability
 - Data and code availability
- EXPERIMENTAL MODEL AND SUBJECT DETAILS
 - Cell culture
- METHOD DETAILS
 - Protein expression and purification
 - RaPID Selection
 - Comparison selection
 - Bio-layer interferometry (BLI)
 - Adenylyl cyclase activity assay
 - $G_{\alpha s}$ /adenylyl cyclase interaction assay
 - The cADDIS cAMP assay
 - Steady-state GTPase assay
 - GDP dissociation assay
 - GTP γ S binding assay
 - FRET based G_{α} /G $\beta\gamma$ interaction assay
 - Crystallization
 - Data collection and structure determination
 - BRET2 based G_{α} G $\beta\gamma$ interaction assay
 - Whole-cell voltage-clamp recordings
 - Chemical stability assay
 - Chemical synthesis
 - Characterization data for cyclic peptides
- QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.cell.2022.09.019>.

ACKNOWLEDGMENTS

We would like to thank the staff at A.L.S. beamline 8.2.1; Drs. X. Wan and X. Liu for assistance in X-ray data processing and structure refinement; Drs. A. Inoue and J. Kritzer for sharing the cells; Dr. A. Born for sharing Gαq; Drs. R. MacKinnon, B. Roth, R. Strachan, R. Olsen for their help with cell-based assays. Dr. K. Zhang, Dr. M. Garcia-Marcos, J. Stevenson, D. Wassarman, and A. Ecker for helpful comments. This work was supported by the Howard Hughes Medical Institute (K.M.S.); NIH grants 5R01CA244550 (K.M.S.), DA010711, DA012864, and MH120212 (M.v.Z.); Japan Agency for Medical Research and Development (AMED), Platform Project for Supporting Drug Discovery and Life Science Research (Basis for Supporting Innovative Drug Discovery and Life Science Research) under JP19a.m.0101090 and the Japan Society for the Promotion of Science (JSPS) Grant-in-Aid for Specially Promoted Research JP20H05618 (H.S.). S.A.D. is a UCSF Discovery fellow and a Fletcher Jones Fellow.; Q.H., K.K.T., and Z.Z. are Damon Runyon fellows supported by the Damon Runyon Cancer Research Foundation (DRG-[2229-15]; DRG-[2387-30]; DRG-[2281-17]); E.E.B. is supported by an NIH National Research Service Award (NRSA) Postdoctoral Fellowship (F32CA260118).

AUTHOR CONTRIBUTIONS

K.M.S. and H.S. conceived the project; S.A.D., Q.H., R.G., E.E.B., K.K.T., K.M.S., and H.S. designed the experiments. R.G. and H.P. did RaPID selection; S.A.D., R.G., H.P., and Z.Z. synthesized the cyclic peptides; S.A.D. and Q.H. performed biochemical characterization of the cyclic peptides; Q.H. crystallized the GN13/Gαs and GD20/Gαs complexes and determined the structures; E.E.B. did the cADDIS assay; K.K.T. did the whole-cell voltage-clamp recordings; S.A.D. did all other cell-based assays; and S.A.D. and K.M.S. wrote the manuscript with the contribution from other authors.

DECLARATION OF INTERESTS

S.A.D., Q.H., R.W., H.P., H.S., and K.M.S. are inventors on patent applications jointly owned by University of Tokyo and UCSF. S.A.D., Q.H., R.W., H.P., H.S., and K.M.S. own shares in G-Protein Therapeutics, a subsidiary of Bridge Bio. K.M.S. has consulting agreements with the following companies, which involve monetary and/or stock compensation: Revolution Medicines, Black Diamond Therapeutics, BridGene Biosciences, Denali Therapeutics, Dice Molecules, eFFECTOR Therapeutics, Erasca, Genentech/Roche, G-Protein Therapeutics, Janssen Pharmaceuticals, Kumquat Biosciences, Kura Oncology, Mitokinin, Nested, Type6 Therapeutics, Venthera, Wellspring Biosciences (Araxes Pharma), Turning Point, Ikena, Initial Therapeutics, Vevo, Rezo, and BioTheryX.

Received: March 18, 2022

Revised: August 8, 2022

Accepted: September 7, 2022

Published: September 27, 2022

REFERENCES

Adams, P.D., Afonine, P.v., Bunkóczi, G., Chen, V.B., Davis, I.W., Echols, N., Headd, J.J., Hung, L.W., Kapral, G.J., Grosse-Kunstleve, R.W., et al. (2010). PHENIX: A comprehensive Python-based system for macromolecular structure solution. *Acta Crystallogr. D Biol. Crystallogr.* 66, 213–221. <https://doi.org/10.1107/S0907444909052925>.

Alessi, D.R., and Sammler, E. (2018). LRRK2 kinase in Parkinson's disease. *Science* 360, 36–37. <https://doi.org/10.1126/science.aar5683>.

Austin, R.J., Ja, W.W., and Roberts, R.W. (2008). Evolution of Class-Specific Peptides Targeting a Hot Spot of the Gαs Subunit. *J. Mol. Biol.* 377, 1406–1418. <https://doi.org/10.1016/j.jmb.2008.01.032>.

Bechara, C., and Sagan, S. (2013). Cell-penetrating peptides: 20 years later, where do we stand? *FEBS Lett.* 587, 1693–1702. <https://doi.org/10.1016/j.febslet.2013.04.031>.

Bonacci, T.M., Mathews, J.L., Yuan, C., Lehmann, D.M., Malik, S., Wu, D., Font, J.L., Bidlack, J.M., and Smrcka, A.V. (2006). Differential Targeting of Gβγ-Subunit Signaling with Small Molecules. *Science* 312, 443–446. <https://doi.org/10.1126/science.1120378>.

Canon, J., Rex, K., Saiki, A.Y., Mohr, C., Cooke, K., Bagal, D., Gaida, K., Holt, T., Knutson, C.G., Koppada, N., et al. (2019). The clinical KRAS(G12C) inhibitor AMG 510 drives anti-tumour immunity. *Nature* 575, 217–223. <https://doi.org/10.1038/s41586-019-1694-1>.

Chang, F.H., and Bourne, H.R. (1989). Cholera toxin induces cAMP-independent degradation of Gs. *J. Biol. Chem.* 264, 5352–5357. [https://doi.org/10.1016/S0021-9258\(18\)83552-9](https://doi.org/10.1016/S0021-9258(18)83552-9).

Chen, Z., Singer, W.D., Sternweis, P.C., and Sprang, S.R. (2005). Structure of the p115RhoGEF rgRGS domain-Gα13/11 chimera complex suggests convergent evolution of a GTPase activator. *Nat. Struct. Mol. Biol.* 12, 191–197. <https://doi.org/10.1038/nsmb888>.

Chen, Z., Singer, W.D., Danesh, S.M., Sternweis, P.C., and Sprang, S.R. (2008). Recognition of the activated states of Gα13 by the rgRGS domain of PDZRhoGEF. *Structure* 16, 1532–1543. <https://doi.org/10.1016/j.str.2008.07.009>.

Dougherty, P.G., Sahni, A., and Pei, D. (2019). Understanding Cell Penetration of Cyclic Peptides. *Chem. Rev.* 119, 10241–10287. <https://doi.org/10.1021/acs.chemrev.9b00008>.

Dror, R.O., Mildorf, T.J., Hilger, D., Manglik, A., Borhani, D.W., Arlow, D.H., Philippsen, A., Villanueva, N., Yang, Z., Lerch, M.T., et al. (2015). Structural basis for nucleotide exchange in heterotrimeric G proteins. *Science* 348, 1361–1365. <https://doi.org/10.1126/science.aaa5264>.

Emsley, P., Lohkamp, B., Scott, W.G., and Cowtan, K. (2010). Features and development of Coot. *Acta Crystallogr. D Biol. Crystallogr.* 66, 486–501. <https://doi.org/10.1107/S0907444910007493>.

Evans, P. (2006). Scaling and assessment of data quality. *Acta Crystallogr. D Biol. Crystallogr.* 62, 72–82. <https://doi.org/10.1107/S0907444905036693>.

Ghosh, P., Rangamani, P., and Kufareva, I. (2017). The GAPs, GEFs, GDIs and...now, GEMs: New kids on the heterotrimeric G protein signaling block. *Cell Cycle* 16, 607–612. <https://doi.org/10.1080/15384101.2017.1282584>.

Goto, Y., Katoh, T., and Suga, H. (2011). Flexizymes for genetic code reprogramming. *Nat. Protoc.* 6, 779–790. <https://doi.org/10.1038/nprot.2011.331>.

Gulati, S., Jin, H., Masuho, I., Orban, T., Cai, Y., Pardon, E., Martemyanov, K.A., Kiser, P.D., Stewart, P.L., Ford, C.P., et al. (2018). Targeting G protein-coupled receptor signaling at the G protein level with a selective nanobody inhibitor. *Nat. Commun.* 9, 1996. <https://doi.org/10.1038/s41467-018-04432-0>.

Hallin, J., Engstrom, L.D., Hargis, L., Calinisan, A., Aranda, R., Briere, D.M., Sudhakar, N., Bowcut, V., Baer, B.R., Ballard, J.A., et al. (2020). The KRASG12C inhibitor MRTX849 provides insight toward therapeutic susceptibility of KRAS-mutant cancers in mouse models and patients. *Cancer Discov.* 10, 54–71. <https://doi.org/10.1158/2159-8290.cd-19-1167>.

Hu, Q., and Shokat, K.M. (2018). Disease-Causing Mutations in the G Protein Gαs Subvert the Roles of GDP and GTP. *Cell* 173, 1254–1264.e11. <https://doi.org/10.1016/j.cell.2018.03.018>.

Ja, W.W., and Roberts, R.W. (2004). In vitro selection of state-specific peptide modulators of G protein signaling using mRNA display. *Biochemistry* 43, 9265–9275. <https://doi.org/10.1021/bi0498398>.

Ja, W.W., Wiser, O., Austin, R.J., Jan, L.Y., and Roberts, R.W. (2006). Turning G proteins on and off using peptide ligands. *ACS Chem. Biol.* 1, 570–574. <https://doi.org/10.1021/cb600345k>.

Johnston, C.A., Willard, F.S., Jezyk, M.R., Fredericks, Z., Bodor, E.T., Jones, M.B., Blaesius, R., Watts, V.J., Harden, T.K., Sondek, J., et al. (2005). Structure of Gαi1 bound to a GDP-selective peptide provides insight into guanine

- p nucleotide exchange.
- Structure*
- 13, 1069–1080.
- <https://doi.org/10.1016/j.str.2005.04.007>
- .
- Johnston, C.A., Ramer, J.K., Blaesius, R., Fredericks, Z., Watts, V.J., and Siderovski, D.P. (2005). A bifunctional $G\alpha i/G\alpha s$ modulatory peptide that attenuates adenylyl cyclase activity. *FEBS (Fed. Eur. Biochem. Soc.) Lett.* 579, 5746–5750. <https://doi.org/10.1016/j.febslet.2005.09.059>.
- Johnston, C.A., Lobanova, E.S., Shavkunov, A.S., Low, J., Ramer, J.K., Blaesius, R., Fredericks, Z., Willard, F.S., Kuhlman, B., Arshavsky, V.Y., and Siderovski, D.P. (2006). Minimal Determinants for Binding Activated $G\alpha$ from the Structure of a $G\alpha i1$ –Peptide Dimer. *Biochemistry* 45, 11390–11400. <https://doi.org/10.1021/bi0613832>.
- Kaur, H., Harris, P.W.R., Little, P.J., and Brimble, M.A. (2015). Total synthesis of the cyclic depsipeptide YM-280193, a platelet aggregation inhibitor. *Org. Lett.* 17, 492–495. <https://doi.org/10.1021/ol503507g>.
- Kreutz, B., Yau, D.M., Nance, M.R., Tanabe, S., Tesmer, J.J.G., and Kozasa, T. (2006). A new approach to producing functional $G\alpha$ subunits yields the activated and deactivated structures of $G\alpha 12/13$ proteins. *Biochemistry* 45, 167–174. <https://doi.org/10.1021/bi051729t>.
- Lambright, D.G., Noel, J.P., Hamm, H.E., and Sigler, P.B. (1994). Structural determinants for activation of the α -subunit of a heterotrimeric G protein. *Nature* 369, 621–628. <https://doi.org/10.1038/369621a0>.
- Liu, X., Xu, X., Hilger, D., Aschauer, P., Tiemann, J.K., Du, Y., Liu, H., Hirata, K., Sun, X., Guixà-González, R., et al. (2019). Structural insights into the process of GPCR-G protein complex formation. *Cell* 177, 1243–1251.e12. <https://doi.org/10.1016/j.cell.2019.04.021>.
- Manglik, A., Kobilka, B.K., and Steyaert, J. (2017). Nanobodies to study G protein-coupled receptor structure and function. *Annu. Rev. Pharmacol. Toxicol.* 57, 19–37. <https://doi.org/10.1146/annurev-pharmtox-010716-104710>.
- Maziarz, M., Park, J.C., Leyme, A., Marivin, A., Garcia-Lopez, A., Patel, P.P., and Garcia-Marcos, M. (2020). Revealing the Activity of Trimeric G-proteins in Live Cells with a Versatile Biosensor Design. *Cell* 182, 770–785.e16. <https://doi.org/10.1016/j.cell.2020.06.020>.
- McCoy, A.J., Grosse-Kunstleve, R.W., Adams, P.D., Winn, M.D., Storoni, L.C., and Read, R.J. (2007). Phaser crystallographic software. *J. Appl. Crystallogr.* 40, 658–674. <https://doi.org/10.1107/s0021889807021206>.
- Morimoto, J., Hayashi, Y., and Suga, H. (2012). Discovery of macrocyclic peptides armed with a mechanism-based warhead: Isoform-selective inhibition of human deacetylase SIRT2. *Angew. Chem. Int. Ed. Engl.* 124, 3479–3483. <https://doi.org/10.1002/ange.201108118>.
- Murakami, H., Saito, H., and Suga, H. (2003). A Versatile tRNA Aminoacylation Catalyst Based on RNA. *Chem Biol* 10, 655–662. [https://doi.org/10.1016/s1074-5521\(03\)00145-5](https://doi.org/10.1016/s1074-5521(03)00145-5).
- Murakami, H., Ohta, A., Ashigai, H., and Suga, H. (2006). A highly flexible tRNA acylation method for non-natural polypeptide synthesis. *Nat. Methods* 3, 357–359. <https://doi.org/10.1038/nmeth877>.
- Neklesa, T.K., Tae, H.S., Schneekloth, A.R., Stulberg, M.J., Corson, T.W., Sundberg, T.B., Raina, K., Holley, S.A., and Crews, C.M. (2011). Small-molecule hydrophobic tagging-induced degradation of HaloTag fusion proteins. *Nat. Chem. Biol.* 7, 538–543. <https://doi.org/10.1038/nchembio.597>.
- Nishimura, A., Kitano, K., Takasaki, J., Taniguchi, M., Mizuno, N., Tago, K., Hakoshima, T., Itoh, H., and Gilman, A.G. (2010). Structural basis for the specific inhibition of heterotrimeric Gq protein by a small molecule. *Proc. Natl. Acad. Sci. USA* 107, 13666–13671. <https://doi.org/10.1073/pnas.1003553107>.
- O’Hayre, M., Vázquez-Prado, J., Kufareva, I., Stawiski, E.W., Handel, T.M., Shagiri, S., and Gutkind, J.S. (2013). The emerging mutational landscape of G proteins and G-protein-coupled receptors in cancer. *Nat. Rev. Cancer* 13, 412–424. <https://doi.org/10.1038/nrc3521>.
- Olsen, R.H.J., DiBerto, J.F., English, J.G., Glaudin, A.M., Krumm, B.E., Slocum, S.T., Che, T., Gavin, A.C., McCovry, J.D., Roth, B.L., and Strachan, R.T. (2020). TRUPATH, an open-source biosensor platform for interrogating the GPCR transducerome. *Nat. Chem. Biol.* 16, 841–849. <https://doi.org/10.1038/s41589-020-0535-8>.
- Otwinowski, Z., and Minor, W. (1997). [20] Processing of X-ray diffraction data collected in oscillation mode. *Methods Enzymol.* 276 (Academic press), pp. 307–326.
- Passioura, T., and Suga, H. (2017). A RaPID way to discover nonstandard macrocyclic peptide modulators of drug targets. *Chem. Commun. (Camb.)* 53, 1931–1940. <https://doi.org/10.1039/c6cc06951g>.
- Peraro, L., Deprey, K.L., Moser, M.K., Zou, Z., Ball, H.L., Levine, B., and Kritzer, J.A. (2018). Cell Penetration Profiling Using the Chloroalkane Penetration Assay. *J. Am. Chem. Soc.* 140, 11360–11369. <https://doi.org/10.1021/jacs.8b06144>.
- Pettersen, E.F., Goddard, T.D., Huang, C.C., Couch, G.S., Greenblatt, D.M., Meng, E.C., and Ferrin, T.E. (2004). UCSF Chimera—a visualization system for exploratory research and analysis. *J. Comput. Chem.* 25, 1605–1612. <https://doi.org/10.1002/jcc.20084>.
- Prior, I.A., Lewis, P.D., and Mattos, C. (2012). A comprehensive survey of ras mutations in cancer. *Cancer Res.* 72, 2457–2467. <https://doi.org/10.1158/0008-5472.can-11-2612>.
- Ramaswamy, K., Saito, H., Murakami, H., Shiba, K., and Suga, H. (2004). Designer ribozymes: Programming the tRNA specificity into flexizyme. *J. Am. Chem. Soc.* 126, 11454–11455. <https://doi.org/10.1021/ja046843y>.
- Rasmussen, S.G.F., DeVree, B.T., Zou, Y., Kruse, A.C., Chung, K.Y., Kobilka, T.S., Thian, F.S., Chae, P.S., Pardon, E., Calinski, D., et al. (2011). Crystal structure of the $\beta 2$ adrenergic receptor-Gs protein complex. *Nature* 477, 549–555. <https://doi.org/10.1038/nature10361>.
- Seifert, R., Wenzel-Seifert, K., Lee, T.W., Gether, U., Sanders-Bush, E., and Kobilka, B.K. (1998). Different effects of Gaa splice variants on $\beta 2$ -adrenoreceptor-mediated The $\beta 2$ -adrenoreceptor coupled to the long splice variant of Gaa has properties of a constitutively active receptor. *J. Biol. Chem.* 273, 5109–5116.
- Slep, K.C., Kercher, M.A., He, W., Cowan, C.W., Wensel, T.G., and Sigler, P.B. (2001). Structural determinants for regulation of phosphodiesterase by a G protein at 2.0 Å. *Nature* 409, 1071–1077. <https://doi.org/10.1038/35059138>.
- Sohrabi, C., Foster, A., and Tavassoli, A. (2020). Methods for generating and screening libraries of genetically encoded cyclic peptides in drug discovery. *Nat. Rev. Chem* 4, 90–101. <https://doi.org/10.1038/s41570-019-0159-2>.
- Stallaert, W., van der Westhuizen, E.T., Schönegge, A.M., Plouffe, B., Hogue, M., Lukashova, V., Inoue, A., Ishida, S., Aoki, J., le Gouill, C., and Bouvier, M. (2017). Purinergic receptor transactivation by the $\beta 2$ -adrenergic receptor increases intracellular Ca^{2+} in nonexcitable cells. *Mol. Pharmacol.* 91, 533–544. <https://doi.org/10.1124/mol.116.106419>.
- Sunahara, R.K., Tesmer, J.J.G., Gilman, A.G., and Sprang, S.R. (1997). Crystal structure of the adenylyl cyclase activator $G\alpha s$. *Science* 278, 1943–1947. <https://doi.org/10.1126/science.278.5345.1943>.
- Syrovatkin, V., Alegre, K.O., Dey, R., and Huang, X.Y. (2016). Regulation, signaling, and physiological functions of G-proteins. *J. Mol. Biol.* 428, 3850–3868. <https://doi.org/10.1016/j.jmb.2016.08.002>.
- Takasaki, J., Saito, T., Taniguchi, M., Kawasaki, T., Moritani, Y., Hayashi, K., and Kobori, M. (2004). A novel $G\alpha q/11$ -selective inhibitor. *J. Biol. Chem.* 279, 47438–47445. <https://doi.org/10.1074/jbc.m408846200>.
- Taylor, V.G., Bommarito, P.A., and Tesmer, J.J. (2016). Structure of the regulator of G protein signaling 8 (RGS8)- $G\alpha q$ complex: molecular basis for $G\alpha$ selectivity. *J. Biol. Chem.* 291, 5138–5145. <https://doi.org/10.1074/jbc.m115.712075>.
- Tesmer, J.J.G., Sunahara, R.K., Gilman, A.G., and Sprang, S.R. (1997). Crystal Structure of the Catalytic Domains of Adenylyl Cyclase in a Complex with $G\alpha s$ GTP γ S. *Science* 278, 1907–1916. <https://doi.org/10.1126/science.278.5345.1907>.
- Tesmer, V.M., Kawano, T., Shankaranarayanan, A., Kozasa, T., and Tesmer, J.J.G. (2005). Snapshot of Activated G Proteins at the Membrane: The $G\alpha q$ -GRK2-G $\beta\gamma$ Complex. *Science* 310, 1686–1690. <https://doi.org/10.1126/science.1118890>.

- Touhara, K.K., and MacKinnon, R. (2018). Molecular basis of signaling specificity between GIRK channels and GPCRs. *Elife* 7, e42908. <https://doi.org/10.7554/elife.42908>.
- Wall, M.A., Coleman, D.E., Lee, E., Iñiguez-Lluhi, J.A., Posner, B.A., Gilman, A.G., and Sprang, S.R. (1995). The Structure of the G Protein Heterotrimer G α 1 β 1 γ 2. *Cell* 83, 1047–1058. [https://doi.org/10.1016/0092-8674\(95\)90220-1](https://doi.org/10.1016/0092-8674(95)90220-1).
- Weis, W.I., and Kobilka, B.K. (2018). The Molecular Basis of G Protein-Coupled Receptor Activation. *Annu. Rev. Biochem.* 87, 897–919. <https://doi.org/10.1146/annurev-biochem-060614-033910>.
- Wilson, G., Sim, J., McLean, C., Giannandrea, M., Galea, C., Riseley, J., Stephenson, S., Fitzpatrick, E., Haas, S., Pope, K., et al. (2014). Mutations in RAB39B cause X-linked intellectual disability and early-onset parkinson disease with α -synuclein pathology. *Am. J. Hum. Genet.* 95, 729–735. <https://doi.org/10.1016/j.ajhg.2014.10.015>.
- Winn, M.D., Ballard, C.C., Cowtan, K.D., Dodson, E.J., Emsley, P., Evans, P.R., Keegan, R.M., Krissinel, E.B., Leslie, A.G.W., McCoy, A., et al. (2011). Overview of the CCP4 suite and current developments. *Acta Crystallogr. D Biol. Crystallogr.* 67, 235–242. <https://doi.org/10.1107/s0907444910045749>.
- Xiao, H., Murakami, H., Suga, H., and Ferré-D'Amaré, A.R. (2008). Structural basis of specific tRNA aminoacylation by a small in vitro selected ribozyme. *Nature* 454, 358–361. <https://doi.org/10.1038/nature07033>.
- Xiong, X.F., Zhang, H., Underwood, C.R., Harpsøe, K., Gardella, T.J., Wöldike, M.F., Mannstadt, M., Gloriam, D.E., Bräuner-Osborne, H., and Strømgaard, K. (2016). Total synthesis and structure-activity relationship studies of a series of selective G protein inhibitors. *Nat. Chem.* 8, 1035–1041. <https://doi.org/10.1038/nchem.2577>.
- Yamagishi, Y., Shoji, I., Miyagawa, S., Kawakami, T., Katoh, T., Goto, Y., and Suga, H. (2011). Natural product-like macrocyclic N-methyl-peptide inhibitors against a ubiquitin ligase uncovered from a ribosome-expressed de novo library. *Chem Biol* 18, 1562–1570. <https://doi.org/10.1016/j.chembiol.2011.09.013>.
- Zachary, I., Masters, S.B., and Bourne, H.R. (1990). Increased mitogenic responsiveness of Swiss 3T3 cells expressing constitutively active Gs α . *Biochem. Biophys. Res. Commun.* 168, 1184–1193. [https://doi.org/10.1016/0006-291x\(90\)91154-k](https://doi.org/10.1016/0006-291x(90)91154-k).
- Zhang, H., Xiong, X.F., Boesgaard, M.W., Underwood, C.R., Bräuner-Osborne, H., and Strømgaard, K. (2017). Structure–Activity Relationship Studies of the Cyclic Depsipeptide Natural Product YM-254890, Targeting the Gq Protein. *ChemMedChem* 12, 830–834. <https://doi.org/10.1002/cmdc.201700155>.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Bacterial and virus strains		
<i>Escherichia coli</i> BL21(DE3)	Invitrogen	Cat# C600003
MAX Efficiency DH10Bac Competent Cells	Thermo Fisher Scientific	Cat# 10,361,012
Chemicals, peptides, and recombinant proteins		
GDP	Sigma-Aldrich	Cat# G7127-100MG
GTP	Sigma-Aldrich	Cat# 11,140,957,001
ATP	Discoverx	Cat# 90-0099
Guanosine 5'-[β , γ -imido]triphosphate (GNP, GppNHp)	Axorra	Cat# JBS-NU-401-50
100X GTP γ S, 10mM	EMD Millipore	Cat# 20-176
Guanosine 5'-Diphosphate, Trisodium Salt, [8,50-3H]-, Specific Activity: 25-50Ci (0.925-1.85TBq)/mMole, 250 μ Ci (9.25MBq)	Perkin-Elmer	Cat# NET966250UC
GTP, [γ -32P]- 6000 Ci/mmol 10 mCi/ml Lead, 250 μ Ci	Perkin-Elmer	Cat# NEG004Z250UC
GTP γ S, [35S]- 1250 Ci/mmol, 12.5 mCi/ml, 250 μ Ci	Perkin-Elmer	Cat# NEG030H250UC
Forskolin	Cayman Chemical Company	Cat# 11,018: 50 mg
Isoproterenol Hydrochloride	TCI	Cat# I0260
Acetylcholine Chloride	Selleckchem	Cat# S1805
Activated Charcoal Norit	Sigma-Aldrich	Cat# 53,663-250G
Cytoscint-ES liquid scintillation cocktail	MP Biomedicals	Cat# 0,188,245,301
Acrylonitrile	Sigma-Aldrich	Cat# 110,213-5ML
cOmplete Protease Inhibitor Cocktail	Sigma-Aldrich	Cat# 5,056,489,001
Carbenicillin	Goldbio	Cat# C-103-100
Kanamycin	Goldbio	Cat# K-120-25
IPTG	Goldbio	Cat# I2481C100
DTT	Goldbio	Cat# DTT10
Biotin	Sigma-Aldrich	Cat# B4501-5G
M-MLV reverse transcriptase	Promega	Cat# 3683
acetylated BSA	Nacalai Tesque	Cat# 01,278-44
TrypLE™ Express Enzyme (1X), no phenol red	Fisher Scientific	Cat# 12,604,013
PBS, pH 7.4	Thermo Fisher Scientific	Cat# 10,010,049
3-Isobutyl-1-methylxanthine, BioUltra, $\geq 99\%$ (IBMX)	Sigma-Aldrich	Cat# I7018
DMSO sterile filtered	Sigma-Aldrich	Cat# D2650
ct-TAMRA	Promega	Cat# G8251
Coelenterazine-400a (Nanolight Technology)	Prolume Ltd	Cat# 340-1
BSA, Fraction V, low Heavy Metals	EMD Millipore	Cat# 12,659-100GM
Critical commercial assays		
LANCE Ultra cAMP Detection Kit	Perkin-Elmer	Cat# TRF0263
GTPase Colorimetric Assay Kit 480 Tests	Innova Biosciences	Cat# 602-0121
Pierce™ BCA® Protein Assay Kits and Reagents, Thermo Scientific, BCA	Fisher Scientific	Cat# PI23227

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Streptavidin XL665	Cisbio	Cat# 610SAXLF
Anti-6His-Tb cryptate	Cisbio	Cat# 61HI2TLF
Green Up cADDIS cAMP Assay Kit	Montana Molecular	Cat# U0200G

Deposited data

GppNHP-bound G α s in complex with the cyclic peptide inhibitor GN13	This paper	PDB: 7BPH
GDP-bound G α s in complex with the cyclic peptide inhibitor GD20	This paper	PDB: 7E5E
GTP γ S-bound G α s	Sunahara et al. (1997)	PDB: 1AZT
GTP γ S-bound G α s in complex with adenylyl cyclase	Tesmer et al. (1997)	PDB: 1AZS
GDP-bound G α s in complex with G $\beta\gamma$	Liu et al. (2019)	PDB: 6EG8
GDP•AlF $_4^-$ -bound G α 13 in complex with PRG rgRGS domain	Chen et al. (2008)	PDB: 3CX7
GDP•AlF $_4^-$ -bound G α q in complex with RGS8	Taylor et al. (2016)	PDB: 5DO9
GDP•AlF $_4^-$ -bound G α i in complex with KB1573	Johnston et al. (2006)	PDB: 2G83
GDP-bound G α 13 in complex with PRG rgRGS domain	Chen et al. (2008)	PDB: 3CX6
GDP-bound G α q in complex with G $\beta\gamma$ and YM-254890	Nishimura et al. (2010)	PDB: 3AH8
GDP-bound G α i in complex with G $\beta\gamma$	Wall et al. (1995)	PDB: 1GP2
GDP•AlF $_4^-$ -bound G α (t/i) in complex with RGS9 and PDE γ	Slep et al. (2001)	PDB: 1FQJ
G α s/G $\beta\gamma$ / β 2AR/Nb35 complex	Rasmussen et al. (2011)	PDB: 3SN6

Experimental models: Cell lines

Sf9 cells	Thermo Fisher Scientific	Cat# 12,659,017
Halo-Tag-GFP-Mito expressing HeLa cells	J. Kritzer (Tufts University) (Peraro et al., 2018)	N/A
HEK293 cells	ATCC	Cat# CRL-1573
(Parent) HEK293 cells	A. Inoue (Tohoku University) (Stallaert et al., 2017)	N/A
GNAS KO HEK293 cells (CL4)	A. Inoue (Tohoku University) (Stallaert et al., 2017)	N/A

Recombinant DNA

G α s(WT) cloned into a modified pET15b vector	(Hu and Shokat, 2018)	N/A
G α s(Q227L) cloned into a modified pET15b vector	This study	N/A
Human ADCY2 (residues 871-1082) cloned into a modified pET15b vector	(Hu and Shokat, 2018)	N/A
Mouse ADCY5(D628E/S645R) (residues 443-659) cloned into a pET29b vector	(Hu and Shokat, 2018)	N/A
Human GNB1(WT) and GNG2(C68S) cloned into a modified pFastBac Dual vector	(Hu and Shokat, 2018)	N/A
Avi-G α s(WT) cloned into a modified pET15b vector	This study	N/A
Avi-G α i1(WT) cloned into a modified pET15b vector	This study	N/A

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Avi-G α s(S275L) cloned into a modified pET15b vector	This study	N/A
pcDNA3 G α s(WT)-HA	This study	N/A
pcDNA3 G α s(S275L)-HA	This study	N/A
pcDNA3 G α si1(WT)-EE-tagged	G. Peng (UCSF)	N/A
SSF- β 2AR	B. Barsi-Rhyne (UCSF)	N/A
pCEH-Sero-SNAP-hM2R	R. Mackinnon (The Rockefeller University)	N/A
G β 1-C Venus	R. Mackinnon (The Rockefeller University)	N/A
G γ 2-N Venus	R. Mackinnon (The Rockefeller University)	N/A
GIRK4-NLuc	R. Mackinnon (The Rockefeller University)	N/A
pcDNA3.1-Beta1	Olsen et al. (2020)	Addgene plasmid # 140,987
pcDNA3.1-GGamma1-GFP2	Olsen et al. (2020)	Addgene plasmid # 140,989
pcDNA3.1-GGamma2-GFP2	This paper	N/A
pcDNA5/FRT/TO-GA α 1-RLuc8	Olsen et al. (2020)	Addgene plasmid # 140,973
pcDNA5/FRT/TO-GA α phasS-RLuc8	Olsen et al. (2020)	Addgene plasmid # 140,980
Software and algorithms		
GraphPad Prism	GraphPad Software	https://www.graphpad.com/scientific-software/prism/
CCP4	Winn et al. (2011)	http://www.ccp4.ac.uk/
Phenix	Adams et al. (2010)	https://www.phenix-online.org/
Coot	Emsley et al. (2010)	https://www2.mrc-lmb.cam.ac.uk/%20personal/pemsley/coot/
Excel	Microsoft	https://www.microsoft.com/en-us/
Word	Microsoft	https://www.microsoft.com/en-us/
Illustrator	Adobe	https://www.adobe.com/products/illustrator.html
Pymol	The PyMOL Molecular Graphics System, Version 1.8 Schrödinger, LLC.	https://pymol.org/2/
UCSF Chimera	Pettersen et al. (2004)	https://www.cgl.ucsf.edu/chimera
Other		
TALON Metal Affinity Resin	Clontech Laboratories	Cat# 635,503
SOURCE 15Q, 200 mL	GE Healthcare	Cat# 17-0947-05
Superdex 200 Increase 10/300 GL	GE Healthcare	Cat# 28-9909-44
Sephadex G-25	GE Healthcare	Cat# 17,003,201
Dynabeads M280 streptavidin magnetic beads	Thermo Fisher Scientific	Cat# 11206D
Transit 2020	Fisher Scientific	Cat# MIR5404
Lipofectamine 2000 Transfection Reagent	Thermo Fisher Scientific	Cat# 11,668,019
Opti-MEM™ I Reduced Serum Medium	Fisher Scientific	Cat# 31-985-062
Mixed cellulose membrane	EMD Millipore	Cat# GSWP02500
Streptavidin biosensors	Molecular Devices	Cat# 18-5019
Sf-900 III SFM	Thermo Fisher Scientific	Cat# 12,658,027
OptiPlate-384, White Opaque 384-well Microplate	PerkinElmer	Cat# 6,007,290
Greiner 384well, black, flat bottom polypropylene plates	Millipore Sigma	Cat# M1937-32EA
96-well Flat Clear Bottom Black Polystyrene Microplates	Corning	Cat# 3340

(Continued on next page)

Continued

REAGENT or RESOURCE	SOURCE	IDENTIFIER
poly-D-lysine-coated white, clear-bottom 96-well assay plates	Greiner Bio-One	Cat# 655,944
White Adhesive Bottom Seal	Perkin Elmer	Cat# 6,005,199
Dounce tissue grinder set	Millipore Sigma	Cat# D8938-1SET
Spark 20 M plate reader	TECAN	N/A
Synergy H4 Hybrid Microplate Reader	BioTek	N/A
Octet RED384	ForteBio	N/A
LS 6500 Multi-Purpose Scintillation Counter	Beckman Coulter	N/A
Axopatch 200B amplifier	Molecular Devices	N/A
Digidata 1550B digitizer	Molecular Devices	N/A
Sutter P-97 puller	Sutter Instrument Company	N/A
Syro Wave automated peptide synthesizer	Biotage	N/A

RESOURCE AVAILABILITY**Lead contact**

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Kevan M. Shokat (kevan.shokat@ucsf.edu).

Materials availability

Plasmids generated in this study are available from the lead contact. Compounds generated in this study will be available from the lead contact upon execution of a materials transfer agreement. Cells used in this study cannot be sent because they were made using cell lines from other labs.

Data and code availability

- Data X-ray Crystallography data have been deposited at PDB and are publicly available as of the date of publication. The accession number for the crystal structure of GNP-bound G α s in complex with the cyclic peptide inhibitor GN13 reported in this paper is PDB: 7BPH. The accession number for the crystal structure of GDP-bound G α s in complex with the cyclic peptide inhibitor GD20 reported in this paper is PDB: 7E5E.
- This paper does not report original code.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS**Cell culture**

HeLa cells stably expressing the Halo-Tag-GFP-Mito construct were provided by the Kritzer lab ([Peraro et al., 2018](#)). HEK293 cells used for cADDIs were from ATCC (CRL-1573), and were cultured at 37°C, 5% CO₂ in DMEM (Thermo Fisher Scientific, Cat#11965118) supplemented with 10% heat-inactivated FBS (HyClone, cat# SH30910.03c). In all other cell-based assays, wild-type HEK293, GNAS KO HEK293 were provided by the Inoue lab. Wild-type HEK293, GNAS KO HEK293 and HeLa cells were cultured at 37°C, 5% CO₂ in DMEM (Thermo Fisher Scientific, Cat# 11,995,073) supplemented with 10% heat-inactivated FBS (AxiomaBiologix). All the cells are female in origin.

WT G α s, all the mutants of G α s, the C1 domain (residues 442-658, VC1) of human ADCY5 (adenylyl cyclase V) and the C2 domain (residues 871-1082, IIC2) of human ADCY2 (adenylyl cyclase II) were overexpressed in *Escherichia coli* BL21(DE3) cultured in Terrific Broth (TB) Medium. Human GNB1 (G β 1) and GNG2 (G γ 2) were co-expressed in Sf9 insect cells cultured in Sf-900 III SFM medium at 28°C. Human GNB1 (G β 1) and GNG2 (G γ 2) were co-expressed in Sf9 insect cells cultured in Sf-900 III SFM medium at 28°C. Human G α (i13) was expressed in Sf9 insect cells cultured in Sf-900 III SFM medium at 28°C.

METHOD DETAILS**Protein expression and purification**

The following proteins were prepared for the AC assay, the radioactivity assay, and the steady-state GTPase assay:

Standard G α s protein purification (Hu and Shokat, 2018): The gene of residues 7–380 of the short isoform of human G α s (GNAS, accession number in PubMed: NP_536351) with a stop codon at its end was cloned into the NdeI/XhoI site of a modified pET15b vector, in which a Drice cleavage site (AspGluValAsp↓Ala) inserted between the thrombin cleavage site and the NdeI site. The resulting WT protein sequence is as follows:

AHMSKTEDQRNEEKAQREANKKIEKQLQKDKQVYRATHRLLLLGAGESGKSTIVKQMRILHVNGFNGDSEKATKVQDIKNNLKEAIE
TIVAAMSNNLPPVELANPENQFRVDYILSVMNVPDFDFPPEFYEHAKALWEDEGVRACYERSNEYQLIDCAQYFLDKIDVIKQADYVP
SDQDLLRCRVLTSIGFETKFQVDKVNFMFDVGGQRDERRKWIQCFNDVTAIFVAVSSSYNMVIREDNQTNRLQEALNLFKSIWNN
RWLRTISVILFLNKQDLLAEKVLGKSKIEDYFPEFARYTTPEDATPEPGEDPRVTRAKYFIRDEFRLISTASGDGRHYCYPHFTCAVD
TENIRRVFNDCRDIIQRMHLRQYELL

The plasmid was transformed into *E. coli* BL21(DE3). The transformed cells were grown in TB medium supplemented with 50 μ g/mL carbenicillin at 37°C until OD600 reached 0.5, and then cooled to 22°C followed by addition of 40 μ M IPTG. After overnight incubation, the cells were harvested by centrifugation, resuspended in lysis buffer (150 mM NaCl, 25 mM Tris 8.0, 1 mM MgCl₂, protease inhibitor cocktail), and then lysed by a microfluidizer. The cell lysate was centrifuged at 19,000 g for 1 h at 4°C. The supernatant was incubated with TALON Resin at 4°C for 2 h, then the resin was washed by 500 mM NaCl, 25 mM Tris 8.0, 1 mM MgCl₂ and 5 mM imidazole 8.0. G α s was eluted by 25 mM Tris 8.0, 1 mM MgCl₂, 250 mM imidazole 8.0, 10% glycerol and 0.1 mM GDP. After adding 5 mM Dithiothreitol (DTT), the eluate was loaded onto a Source-15Q column. G α s was eluted by a linear gradient from 100% IEC buffer A (25 mM Tris 8.0, 1 mM MgCl₂) to 40% IEC Buffer B (25 mM Tris 8.0, 1 M NaCl, 1 mM MgCl₂). The peak fractions were pooled and supplemented with 5 mM DTT. One-half of peak fractions was mixed with equal volume of GNP exchange buffer (150 mM NaCl, 25 mM HEPES 8.0, 2 mM EDTA, 2 mM GNP, 5 mM DTT) for 2 h, followed by addition of 5 mM MgCl₂. GNP-bound G α s and GDP-bound G α s were concentrated and purified by gel filtration (Superdex 200 increase, 10/30) with SEC buffer (150 mM NaCl, 20 mM HEPES 8.0, 5 mM MgCl₂ and 1 mM EDTA-Na 8.0). The peak fractions were pooled and concentrated for biochemical assay. All mutants of untagged G α s (WT, S275L, R201C, R201H, R201S, and Q227L) were expressed and purified with the same protocol. The adenylyl cyclase C2 domain of human ADCY2 (residues 871–1082, IIC2) was also expressed and purified with the same protocol, except that no GDP was added during purification.

Adenylyl cyclase C1 domain purification: Residues D628 and S645 in the C1 domain (residues 443–659) of mouse ADCY5 (adenylyl cyclase V) were mutated to glutamic acid and arginine, respectively, resulting a sequence that is the same as the C1 domain of human ADCY5 (residues 442–658). The gene of this sequence was cloned into the NdeI/XhoI site of a pET29b vector. The transformed *E. coli* BL21(DE3) cells were cultured in TB medium supplemented with 50 μ g/mL kanamycin at 37°C until OD600 reached 0.5, and then cooled to 22°C followed by addition of 40 μ M IPTG. After incubation at 22°C for 4–5 h, the cells were harvested, lysed. After centrifugation, the supernatant was purified by TALON Resin with the same protocol described above. The eluate was mixed with 5 mM DTT and further purified by gel filtration (Superdex 200 increase, 10/30) with SEC buffer (150 mM NaCl, 20 mM HEPES 8.0, 5 mM MgCl₂ and 1 mM EDTA-Na 8.0).

G $\beta\gamma$ purification: Human G β 1 with a hexahistidine tag at its N terminus and human G γ 2(C68S) were cloned into pFastBac Dual expression vector. The plasmid was transformed into DH10Bac competent cells to generate bacmid DNA, which was then used to generate baculoviruses in Sf9 insect cells. Sf9 cells grown in Sf-900 III SFM medium with a density of 1.8×10^6 cells/mL was infected by the baculoviruses. 48 h later, the cells were harvested by centrifugation, and resuspended in lysis buffer (150 mM NaCl, 25 mM Tris 8.0, 1 mM MgCl₂) supplemented with protease inhibitor cocktail. The cells were disrupted by a microfluidizer. After centrifugation, the supernatant was purified by TALON Resin and gel filtration (Superdex 200 increase, 10/30) with the same buffers used for G α s purification.

The following proteins were prepared for the RaPID selection:

The gene encoding residues 7–380 of the short isoform of human G α s (GNAS, accession number in PubMed: NP_536351) with an Avi tag and a TEV cleavage site at its N-terminus was cloned into the multiple cloning site 1 of the pETDuet vector. The resulting protein sequence is as follows:

MGSSHHHHHSGMSGLNDIFEAQKIEWHESSGENLYFQGMSTEDQRNEEKAQREANKKIEKQLQKDKQVYRATHRLLLLGAGES
GKSTIVKQMRILHVNGFNGDSEKATKVQDIKNNLKEAIEIVAAMSNNLPPVELANPENQFRVDYILSVMNVPDFDFPPEFYEHAKALW
EDEGVRACYERSNEYQLIDCAQYFLDKIDVIKQADYVPSQDLLRCRVLTSIGFETKFQVDKVNFMFDVGGQRDERRKWIQCFNDV
TAIFVAVSSSYNMVIREDNQTNRLQEALNLFKSIWNNRWLRTISVILFLNKQDLLAEKVLGKSKIEDYFPEFARYTTPEDATPEPGED
PRVTRAKYFIRDEFRLISTASGDGRHYCYPHFTCAVDTENIRRVFNDCRDIIQRMHLRQYELL

In the same pETDuet plasmid, the gene encoding BirA (accession number in PubMed: NP_418404.1) was inserted between NdeI and XhoI sites of the multiple cloning site 2. This plasmid was transformed into *E. coli* BL21(DE3). The transformed cells were grown in TB medium supplemented with 50 μ g/mL carbenicillin at 37°C until OD600 reached 0.5, and then cooled to 22°C followed by addition of 40 μ M IPTG. After overnight incubation, 50 μ M biotin was added into the culture for 2 h. The cells were harvested by centrifugation after biotinylation and purified using the standard G α s protein purification protocol.

The following proteins were prepared for the TR-FRET assay and the bio-layer interferometry assay:

The gene of residues 7–380 of the short isoform of human G α s (GNAS, accession number in PubMed: NP_536351) with a stop codon at its end was cloned into the NdeI/XhoI site of a modified pET15b vector, in which a Drice cleavage site (AspGluValAsp↓Ala) and an Avi tag were inserted at the N-terminus. The resulting WT G α s protein sequence after Drice protease cleavage is as follows:

AHMGLNDIFEAQKIEWHESKTEDQRNEEKAQREANKKIEKQLQKDKQVYRATHRLLLLGAGESGKSTIVKQMRILHVNGFNGDSEKA
TKVQDIKNNLKEAETIVAAMSNLVPVELANPENQFRVDYILSVMNVPDFDFPPEFYEHAKALWEDEGVRACYERSNEYQLIDCAQY
FLDKIDVIKQADYVPSDQDLLRCRVLTSGLFETKFQVDKVNFMFDVGGQRDERRKWIQCFNDVTAIIFVASSSYNMVIREDNQNTNR
LQEALNLFKSIWNNRWLRTISVILFLNKQDLLAEKVLGKSKIEDYFPEFARYTTPEDATPEPGEDPRVTRAKYFIRDEFIRISTASGDG
RHICYPHFTCAVDTENIRRVFNDRCRDIIQRMHLRQYELL

The gene of residues 2–354 of human G α i1 (GNAI1, accession number in PubMed: NP_002060.4) with a stop codon at its end was cloned into the NdeI/XhoI site of a modified pET15b vector, in which a Drice cleavage site (AspGluValAsp↓Ala) and an Avi tag were inserted at the N-terminus. The resulting WT G α i1 protein sequence after Drice protease cleavage is as follows:

AHMGLNDIFEAQKIEWHEGCTLSAEDKAAVERSKMIDRNLRDGEKAAREVKLLLLGAGESGKSTIVKQMKIIEAGYSEEECKQYKA
VVYSNTIQSIIAIRAMGRLLKIDFGDSARADDARQLFVLGAAEEGFMTAELAGVIKRLWKDSGVQACFNRSREYQLNDSAAYYLNDLD
RIAQPNYIPTQQDVLRTVRKTTGIVETHFTFKDLHFKMFDVGGQRSEKRWIHCFEVTAIIFCVALSVDYDLVLAEDDEEMNRMHESMK
LFDSICNNKWFDTDSIILFLNKKDLFEKIKKSPLTICYPEYAGSNTYEEAAAYIQCFEDLNKRKDTKEIYTHFTCATDTKNVQFVDAV
TDVIKNNLKDCGLF

The above-mentioned plasmids were transformed into *E. coli* BL21(DE3), respectively. The transformed cells were grown in TB medium supplemented with 50 μ g/mL carbenicillin at 37°C until OD₆₀₀ reached 0.4, and then cooled to 22°C followed by addition of 100 μ M IPTG. After overnight incubation, the cells were harvested by centrifugation, resuspended in lysis buffer (150 mM NaCl, 25 mM Tris 8.0, 1 mM MgCl₂, protease inhibitor cocktail), and then lysed by a microfluidizer. The cell lysate was centrifuged at 14,000 g for 1 h at 4°C. The supernatant was incubated with TALON resin at 4°C for 1 h, then the resin was washed by 500 mM NaCl, 25 mM Tris 8.0, 1 mM MgCl₂ and 5 mM imidazole 8.0. G protein was eluted by 25 mM Tris 8.0, 1 mM MgCl₂, 250 mM imidazole 8.0, 10% glycerol and 0.1 mM GDP. After adding 5 mM Dithiothreitol (DTT), the eluate was incubated with Drice protease at 4°C overnight to remove the hexahistidine tag. Purified BirA (A gift from the Wells lab) and biotin were added at 4°C until LC-MS showed complete biotinylation. G protein was loaded onto a Source-15Q column and eluted by a linear gradient from 100% IEC buffer A (25 mM Tris 8.0, 1 mM MgCl₂) to 40% IEC Buffer B (25 mM Tris 8.0, 1 M NaCl, 1 mM MgCl₂). The peak fractions were pooled, nucleotide exchanged, and supplemented with 5 mM DTT and 0.1 mM nucleotide, and then concentrated and purified by gel filtration (Superdex 200 increase, 10/30) with SEC buffer (150 mM NaCl, 20 mM HEPES 8.0, 5 mM MgCl₂ and 1 mM EDTA-Na 8.0). The peak fractions were pooled and concentrated for biochemical assay. All mutants of G α s (WT, D229A, R231A, R232A, E268A, N271A, K274A, N279A, and R280A) were expressed and purified with the same protocol.

The following proteins were prepared for the bio-layer interferometry assay (Kreutz et al., 2006):

The gene of residues 1–28 of human G α i1 (GNAI1, accession number in PubMed: NP_002060.4) and the gene of residues 47–377 of human G α 13 (GNA13, accession number in PubMed: NP_006563.2) with a stop codon at its end was cloned into the pFastBacHTA vector, in which a Drice cleavage site (AspGluValAsp↓Ala) and an Avi tag were inserted at the N-terminus. The resulting protein sequence after Drice protease cleavage is as follows:

AHMGLNDIFEAQKIEWHEMGCTLSAEDKAAVERSKMIDRNLRDGEKARSARLVKILLGAGESGKSTFLKQMRIIHGQDFDQRRAREEF
RPTIYSNVIKGMRVLVDAREKLHIPWGDNSNQHQGDKMMSFDTRAPMAAQGMVETRVFLQYLPALRALWADSGIQNAYDRRREFQ
LGESVKYFLDNLKLGEPDYIPSQQDILLARRPTKGIHEYDFEIKNVFPKMVDVGGQRSEKRWFEFCDSVTSILFLVSSSEFDQVLM
EDRLTNRLTESLNIFETIVNNRVFSNVSIILFLNKTDLLEEKVQIVSIKDYFLEFEGDPHCLRDVQKFLVECFRNKRDRDQQKPLYHHFTT
AINTENIRLVFRDVKDTILHDNLKQLMLQ

Amplified Avi-G α (i/13) baculovirus stock was generated using the above-mentioned plasmid in Sf9 insect cells. Cells from 2 L of Sf9 culture were harvested 48 h after infection with 15 mL/L of amplified baculovirus stock, resuspended in 100 mL of Lysis Buffer (20 mM HEPES, pH 8.0, 0.1 mM EDTA, 10 mM 2-mercaptoethanol (β ME), 3 mM MgCl₂, 100 mM NaCl, 50 μ M GDP, and protease inhibitor cocktail) and then lysed by microfluidizer. The cell lysate was centrifuged at 19,000 g for 1.5 h at 4°C, after which the supernatants were diluted to a final protein concentration of 5 mg/mL with Buffer A (20 mM HEPES, pH 8.0, 10 mM β ME, 1 mM MgCl₂, 100 mM NaCl, 50 μ M GDP, and 12.5 mM imidazole, pH 8.0) and loaded onto TALON resin equilibrated with Buffer A. The resin was washed with 20 vol of Buffer B (Buffer A containing 0.4 M NaCl and 20 mM imidazole, pH 8.0), and the chimera was eluted in 10 fractions of 1 vol of Buffer C (Buffer A containing 150 mM imidazole, pH 8.0). Peak fractions were supplemented with 10% glycerol. The eluate was treated with Drice and 20 μ L of BirA at 4°C until LC-MS showed complete biotinylation. (Final [MgCl₂] = 10mM, [ATP] = 10mM, [Biotin] = 50 μ M). The eluate was nucleotide exchanged and supplemented with 5 mM DTT and 0.1 mM nucleotide, and then concentrated and purified by gel filtration (Superdex 200 increase, 10/30) with SEC buffer (150 mM NaCl, 20 mM HEPES 8.0, 5 mM MgCl₂ and 1 mM EDTA-Na 8.0). The peak fractions were pooled and concentrated for biochemical assay.

RaPID Selection

Selections were performed with thioether-macrocytic peptide library against biotinylated $G\alpha s$. Thioether-macrocytic peptide libraries were constructed with N-chloroacetyl-D-tyrosine (ClAc^DTyr) as an initiator by using the flexible *in vitro* translation (FIT) system (Goto et al., 2011). The mRNA libraries, ClAc^DTyr-tRNA^{fMet}_{CAU} were prepared as reported (Yamagishi et al., 2011). The mRNA library corresponding for the thioether-macrocytic peptide library was designed to have an AUG initiator codon to incorporate N-chloroacetyl-D-tyrosine (ClAc^DTyr), followed by 8–12 NNK random codons (N = G, C, A or U; K = G or U) to code random proteinogenic amino acids, and then a fixed downstream UGC codon to assign Cys. After *in vitro* translation, a thioether bond formed spontaneously between the N-terminal ClAc group of the initiator ^DTyr residue and the sulfhydryl group of a downstream Cys residue.

In the first round of selection, the initial cyclic peptide library was formed by adding puromycin ligated mRNA library (225 pmol) to a 150 μ L scale flexible *in vitro* translation system, in the presence of 30 μ M of ClAc^DTyr-tRNA^{fMet}_{CAU}. The translation was performed 37°C for 30 min, followed by an extra incubation at 25°C for 12 min. After an addition of 15 μ L of 200 mM EDTA (pH 8.0) solution, the reaction solution was incubated at 37°C for 30 min to facilitate cyclization. Then the library was reverse transcribed by M-MLV reverse transcriptase at 42°C for 1 h and subject to pre-washed Sephadex G-25 columns to remove salts. The desalted solution of peptide-mRNA/cDNA was applied to $G\alpha s$ (positive selection state)-immobilized Dynabeads M280 streptavidin magnetic beads and rotated at 4°C for 1 h in selection buffer (25 mM HEPES pH 7.5, 150 mM NaCl, 1 mM MgCl₂ and 0.05% Tween 20) containing 0.5 mM corresponding nucleotide and 0.1% acetylated BSA. Bead amounts were chosen that the final concentration of $G\alpha s$ protein was 200 nM. This process is referred to as positive selection. The selected peptide-mRNA/cDNAs were isolated from the beads by incubating in 1xPCR reaction buffer heated at 95°C for 5 min. The amount of eluted cDNAs was measured by qPCR. The remaining cDNAs were amplified by PCR, purified and transcribed into mRNAs as a library for the next round of selection.

In the subsequent rounds of selection, ligated mRNA from previous round (7.5 pmol) was added to a 5 μ L scale reprogrammed *in vitro* translation system. This was incubated at 37°C for 30 min and at 25°C for 12 min. Then 1 μ L of 100 mM EDTA (pH 8.0) was added and incubated at 37°C for 30 min. After reverse transcription and subject to pre-washed Sephadex G-25 columns to remove salts, negative selection was performed by adding the desalted solution of peptide-mRNA/cDNA to $G\alpha s$ (negative selection state)-immobilized Dynabeads M280 streptavidin magnetic beads and rotated at 4°C for 30 min in selection buffer containing 0.1% acetylated BSA. This process was repeated several times by removing the supernatant to fresh beads immobilized with $G\alpha s$ (negative selection state). The supernatant from the last negative selection was then added to beads immobilized with the positive selection state of $G\alpha s$ (final conc. 200nM) and rotated at 4°C for 30 min in selection buffer containing 0.5mM corresponding nucleotide and 0.1% acetylated BSA. As described in the first round of selection, the cDNA was quantified with qPCR, amplified with PCR, transcribed and ligated to puromycin. The subsequent selection was repeated for several rounds until a significant enrichment of cDNA was observed for positive selection state. The recovered cDNA was then identified by next generation sequencing (Miseq, Illumina).

Comparison selection

In comparison selection, ligated mRNA (7.5 pmol) from last round selection was added to a 5 μ L scale reprogrammed *in vitro* translation system. After translation, cyclization, reverse transcription and pre-washed with Sephadex G-25 columns, the desalted solution of peptide-mRNA/cDNA library was split equally into three fractions, and perform three paralleled selections with the same amount of blank, GDP-bound $G\alpha s$ -immobilized or GNP-bound $G\alpha s$ -immobilized Dynabeads M280 streptavidin magnetic beads, individually. For each of the paralleled selections, the beads were rotate at 4°C for 30 min, washed three times with selection buffer. The remaining cDNAs were then eluted from the beads, quantified by qPCR, followed by Miseq sequencing. Finally, identified sequences from each paralleled selection were compared by normalization of Miseq abundance of the sequence with the qPCR reads of the paralleled selection.

Bio-layer interferometry (BLI)

BLI experiments were performed using an OctetRED384 instrument from ForteBio. All experiments were performed at 25°C using BLI buffer (10 mM HEPES pH 7.4, 150 mM NaCl, 1mM MgCl₂, 0.05% Tween 20, 0.1% DMSO, 0.2 mM GNP or GDP). Cyclic peptides or $G\alpha$ proteins were diluted to a series of concentrations (Final concentrations were indicated in Figures) in BLI buffer plus 10 μ M Biotin. Assays were conducted in Greiner 384well, black, flat bottom polypropylene plates containing the protein solutions, BLI buffer plus 10 μ M Biotin for dissociation, and serial dilutions of cyclic peptides to be tested.

Biotinylated proteins or cyclic peptides were immobilized on Streptavidin biosensors by dipping sensors into plate wells containing protein solutions at a concentration of 50–150 nM. Protein loading is around 2–3 nm. Cyclic peptide loading is around 0.2–0.3 nm. Sensors loaded with proteins or cyclic peptides were moved and dipped into wells with BLI buffer plus 10 μ M Biotin to block unlabeled Streptavidin. Association-dissociation cycles of were started by moving and dipping sensors to cyclic peptides dilutions and BLI buffer plus 10 μ M Biotin wells alternatively. Association and dissociation times were indicated in the figure legend.

Raw kinetic data collected were processed with the Data Analysis software provided by the manufacturer using single reference subtraction in which buffer-only reference was subtracted (For GN13 analysis). Because GD20 analogs have a low level of background binding, we used a double reference subtraction (buffer-only reference and non-protein-loading reference) method to calculate their kinetics values. The resulting data were analyzed based on a 1:1 binding model from which k_{on} and k_{off} values were obtained and then K_d values were calculated.

Adenylyl cyclase activity assay

Cyclic peptides (4 mM stock in DMSO) were diluted to 4X stocks with a series of concentrations in reaction buffer (1x PBS 7.4, 0.1% BSA). *Gαs* at a concentration of 8.5 mg/mL (about 190 μM) in 20 mM HEPES 8.0, 150 mM NaCl, 5 mM MgCl₂, 1 mM EDTA-Na 8.0 was diluted to 0.5 μM in dilution buffer (1x PBS 7.4, 0.1% BSA, 1 mM EDTA-Na 8.0, 2 mM DTT, 0.1 mM MgCl₂) plus 1 mM GNP (For the GDP-bound R201 mutants, GDP was used in the nucleotide exchange experiments). After incubation at room temperature for 1 h to allow nucleotide exchange, 2.5 μL of 4x *Gαs* dilution was mixed with 1 μL MgCl₂ stock (20 mM MgCl₂, 1x PBS 7.4, 0.1% BSA) in an OptiPlate-384, White Opaque 384-well Microplate to lock *Gαs* in GNP-bound state. 2 μL of 5x AC stock (2 μM VC1, 2 nM IIC2, 150 μM FSK, 1x PBS 7.4, 0.1% BSA) was added, followed by addition of 2.5 μL 4X cyclic peptides stock. Reaction mixture was further incubated at room temperature for 2 h and placed on ice for 5 min. cAMP production was initiated by addition of 2 μL of ATP stock (1 mM ATP, 1x PBS 7.4, 0.1% BSA). The reaction was carried out at 30°C for 10 min in a PCR machine and stopped by heating at 95°C for 3 min. The cAMP concentrations were measured by the LANCE Ultra cAMP kit. Final [cyclic peptide]: 0, 0.39, 0.78, 1.56, 3.12, 6.25, 12.5, 25 μM; Final [*Gαs*]: 125 nM; Final [VC1]: 400 nM; Final [IIC2]: 0.4 nM; Final [FSK]: 30 μM; Final [ATP]: 200 μM. This protocol was used for [Figures 2C, 2S2H, S3A, S3J, and S5J](#).

WT *Gαs* and S275L mutant at a concentration of 8.5 mg/mL (about 190 μM) in 20 mM HEPES 8.0, 150 mM NaCl, 5 mM MgCl₂, 1 mM EDTA-Na 8.0 were diluted to a series of concentrations in dilution buffer (1x PBS 7.4, 0.1% BSA, 1 mM EDTA-Na 8.0, 2 mM DTT, 0.1 mM MgCl₂) plus 1 mM GNP. After incubation at room temperature for 1 h to allow nucleotide exchange, 2.5 μL of 4x each sample was then mixed with 1 μL of MgCl₂ stock (20 mM MgCl₂, 1x PBS 7.4, 0.1% BSA) in an OptiPlate-384, White Opaque 384-well Microplate. 2 μL of AC/Gβγ 5x stock (2 μM VC1, 2 nM IIC2, 150 μM FSK, 1x PBS 7.4, 0.1% BSA, 10 μM Gβ1/γ2(C68S)) was added, followed by addition of 2.5 μL 25 μM GN13 4x stock in 1x PBS 7.4, 0.1% BSA. Reaction mixture was further incubated at room temperature for 2 h and placed on ice for 5 min. cAMP production was initiated by addition of 2 μL of ATP stock (1 mM ATP, 1x PBS 7.4, 0.1% BSA). The reaction was carried out at 30°C for 10 min in a PCR machine and stopped by heating at 95°C for 3 min. The cAMP concentrations were measured by the LANCE Ultra cAMP kit. Final [cyclic peptide]: 6.25 μM; Final [*Gαs*]: 0, 1.37, 4.12, 12.3, 37.0, 111, 333, 1000 nM; Final [VC1]: 400 nM; Final [IIC2]: 0.4 nM; Final [FSK]: 30 μM; Final [Gβ1/γ2(C68S)]: 2 μM; Final [ATP]: 200 μM. This protocol was used for [Figure 3H](#).

Cell membrane preparation: HEK293 cells, *GNAS* KO HEK293 cells were plated two day before transfection at a density of 1M cells per 10cm plate. One plate of *GNAS* KO HEK293 cells was transfected with 4 μg of *GNAS* WT or *GNAS* S275L plasmids. After overnight transfection, cells were lifted with TrypLE, washed, resuspended in stimulation buffer (1x PBS, protease inhibitor cocktail, 5 mM MgCl₂). Cell membranes were disrupted by using the Dounce homogenizer for 25 strokes. Nuclei and unbroken cells were removed by centrifugation for 5 min at 500 g. The supernatant suspension was carefully removed and centrifuged for 30 min at 45K. Cell membranes were suspended in stimulation buffer. The protein concentrations were measured using BCA, and were normalized to 750 μg/mL with stimulation buffer. A final concentration of 0.1% BSA was added into the cell membrane suspension. AC activity assay in cell membranes: 600 μL of cell membrane suspension was mixed with 60 μL of GTP/GDP 20x stock (stock concentration: 10 mM/1 mM). 5.5 μL of the mixture from last step was mixed with 2.5 μL of GN13 4x stocks and incubated at room temperature. After 2 h, membrane/cyclic peptide mixture was transferred on ice for 5 min, followed by the addition of 2 μL of IBMX/ISO/ATP or IBMX/DMSO/ATP 5x stock (5 mM IBMX, 0.2 mM ISO or DMSO, 2.5 mM ATP in stimulation buffer with 0.1% BSA). The reaction was carried out at 30°C for 30 min in a PCR machine and stopped by heating at 95°C for 3 min. The cAMP concentrations were measured by the LANCE Ultra cAMP kit. Final [cyclic peptide]: 0, 0.78, 1.56, 3.12, 6.25, 12.5, 25, 50 μM; Final [membrane]: 375 μg/mL; Final [IBMX]: 1 mM; Final [ISO]: 40 μM; Final [ATP]: 500 μM; Final [GTP]: 500 μM; Final [GDP]: 50 μM. This protocol was used for the HEK293 cell membranes AC assay ([Figures 2F, 3I, and S5K](#)).

cAMP concentrations measurement by the LANCE Ultra cAMP kit: A cAMP standard curve was generated in the same plate using the 50 μM cAMP standard in the kit. Before the measurement, the samples were diluted by stimulation buffer (1x PBS 7.4, 0.1% BSA) to 1/60, 1/120, 1/240 or 1/480 to make sure the cAMP concentrations were in the dynamic range of the cAMP standard curve. 10 μL of each diluted sample was mixed with 5 μL of 4X Eu-cAMP tracer and 5 μL of 4X ULIGHT-anti-cAMP in a white, opaque Optiplate-384 microplate, incubated for 1 h at room temperature, and the time-resolved fluorescence resonance energy transfer (TR-FRET) signals were read on a Spark 20M plate reader. The cAMP standard curve was fitted by the software GraphPad Prism using the following equation in which “Y” is the TR-FRET signal and “X” is the log of cAMP standard concentration (M):

$$Y = \text{Bottom} + (\text{Top} - \text{Bottom}) / (1 + 10^{((\text{LogIC50} - X) * \text{HillSlope}))}$$

After obtained the values of the four parameters “Bottom”, “Top”, “LogIC50” and “HillSlope”, we used this equation to convert the TR-FRET signals of the samples into cAMP production values. The cyclic peptides dose dependent inhibition curves were fitted by the following equation to calculate the IC50 of each cyclic peptide:

$$Y = \text{Bottom} + (\text{Top} - \text{Bottom}) / (1 + 10^{((\text{LogIC50} - X) * \text{HillSlope}))},$$

in which “Y” is the cAMP production value, “X” is the log of cyclic peptide concentration (M).

Gαs/adenylyl cyclase interaction assay

Cyclic peptides, GN13 and others (4 mM stock in DMSO) were diluted to 5X stocks with a series of concentrations in assay buffer (1x PBS 7.4, 0.1% BSA, 2 mM DTT, 2 mM MgCl₂). WT *Gαs* and *Gαs* S275L mutant at a concentration of 4.6 mg/mL (about 100 μM) in

20 mM HEPES 8.0, 150 mM NaCl, 5 mM MgCl₂ were diluted to 4 μM in EDTA GNP buffer (1x PBS 7.4, 0.1% BSA, 2 mM EDTA-Na 8.0, 2 mM DTT, 0.1 mM MgCl₂, 1 mM GNP). After incubation at room temperature for 1 h to allow nucleotide exchange, Gαs dilutions were mixed with equal volume of MgCl₂ stock (3.8 mM MgCl₂, 1x PBS 7.4, 0.1% BSA, 2 mM DTT) to lock Gαs in GNP-bound state. GNP-bound Gαs proteins were then diluted to 500 nM (5X stocks) in assay buffer plus 0.5 mM GNP. In an OptiPlate-384 White Opaque 384-well Microplate, 5X Gαs proteins were mixed with 5X GN13 serial dilution stocks, 5X streptavidin XL665 stock (125 nM), 5X AC stock (VC1: 100 nM, IIC2: 200 nM, FSK 0.5 mM) and 5X anti-6His-Tb cryptate stock (0.26 μg/mL) in assay buffer for 1 h at room temperature. The plate was read on a TECAN Spark 20 M plate reader using the TR-FRET mode with the following parameters: Lag time: 70 μs, Integration time: 500 μs, Read A: Ex 320(25) nm (filter), Em 610(20) nm (filter), Gain 130, Read B: Ex 320(25) nm (filter), Em 665(8) nm (filter), Gain 165. FRET Signal was calculated as the ratio of [Read B]/[Read A]. In Figure 2A, Final [cyclic peptide]: 0, 0.020, 0.039, 0.078, 0.16, 0.31, 0.62, 1.25, 2.5, 5, 10, 20 μM; Final [Gαs]: 100 nM; Final [VC1]: 20 nM; Final [IIC2]: 40 nM; Final [FSK]: 100 μM. In Figure S3L, Final [cyclic peptide]: 0, 0.677, 2.03, 6.10, 18.3, 54.9, 165, 494, 1481, 4444, 13,333, 40,000 nM; Final [Gαs]: 100 nM; Final [VC1]: 20 nM; Final [IIC2]: 40 nM; Final [FSK]: 100 μM.

The cADDIS cAMP assay

Real-time cAMP dynamics were measured using the Green Up cADDIS cAMP biosensor according to the manufacturer's protocol. Briefly, cells were lifted using TrypLE Express and resuspended in media supplemented with the appropriate volume of cADDIS BacMam. Cells were plated into a 96-well plate at a concentration of 50,000 cells per well and incubated overnight. In the case of 24 h drug pretreatment, cADDIS media was replaced with 25 μM drug in DMEM supplemented with 1% dialyzed FBS after 4 h. The next day, plates were washed once with assay buffer (20 mM HEPES pH 7.4, 135 mM NaCl, 5 mM KCl, 0.4 mM MgCl₂, 1.8 mM CaCl₂, 5 mM d-glucose) before a 10-min incubation with DMSO or 25 μM drug in a plate reader pre-warmed to 37°C. Fluorescence was detected using an excitation wavelength of 500 nm and an emission wavelength of 530 nm every 30 s. After a 5-min baseline reading, vehicle or 20 nM isoproterenol were added, and fluorescence was measured for 30 min. A baseline fluorescence (F₀) was calculated for each well by averaging its fluorescence over the 5-min baseline reading, and the fluorescence response at each timepoint was calculated as the change in fluorescence (ΔF = F - F₀) normalized to the baseline (F₀). Each biological replicate represents the average of at least two technical replicates.

Steady-state GTPase assay

WT Gαs (both short and long) was diluted to a 6 μM stock (4X) in GTPase assay buffer (20 mM HEPES 7.5, 150 mM NaCl, 1 mM MgCl₂). The protein was 1:1 (v/v) diluted with 4X cyclic peptide stock (0, 1.56, 3.12, 6.25, 12.5, 25, 50, 100 μM) in GTPase assay buffer, and incubated at 37°C for an hour. The samples were then 1:1 (v/v) diluted with reaction buffer (20 mM HEPES 7.5, 150 mM NaCl, 1 mM MgCl₂, and 1 mM GTP) and incubated at 37°C. After 30, 50, 70, 90 min, 50 μL of the sample was removed to measure the inorganic phosphate (Pi) concentration by PiColorLock Phosphate Detection kit. A standard curve was made using the 0.1 mM Pi stock in the kit. Final [cyclic peptide]: 0, 0.39, 0.78, 1.56, 3.12, 6.25, 12.5, 25 μM; Final [Gαs]: 1.5 μM; Final [GTP]: 500 μM.

GDP dissociation assay

Gα proteins were diluted to 400 nM in the EDTA buffer (20 mM HEPES 7.5, 150 mM NaCl, 1 mM EDTA-Na 8.0, 2 mM DTT). [³H]GDP (1 mCi/mL, 25.2 μM) was added to a final concentration of 1.2 μM, followed by cyclic peptides addition. After incubation at 20°C for 30 min, the same volume of assay buffer (20 μM HEPES-Na 7.5, 150 mM NaCl, 2 mM MgCl₂, and 1 mM GDP) was added to initiate [³H]GDP dissociation. Final [cyclic peptide]: 10 μM; Final [Gα]: 187 nM; Final [GDP]: 500 μM. At various points, 10 μL of the sample was removed and mixed with 390 μL of ice-cold wash buffer (20 mM HEPES 7.5, 150 mM NaCl, 20 mM MgCl₂). The mixture was immediately filtered through a mixed cellulose membrane (25 mm, 0.22 μm) held by a microanalysis filter holder (EMD Millipore). The membrane was washed by ice-cold wash buffer (500 μL x 3), put in a 6-mL plastic vial and air-dried (room temperature 1.5 h). 5 mL of CytoScint-ES Liquid Scintillation Cocktail was added to each vial. After incubation overnight at room temperature, the vial was used for liquid scintillation counting with an LS 6500 Multi-Purpose Scintillation Counter. The GDP dissociation curves were fitted by the software GraphPad Prism using the following equation to calculate the dissociation rates (k_{off}):

$$Y = Y_0 * \exp(-k_{off} * X)$$

in which "Y" is the radioactivity (Counts per minute) of the sample at time "X" (minutes), and Y₀ is the calculated radioactivity of the sample at the time point 0.

GTPγS binding assay

Gα proteins were diluted to 10 μM with dilution buffer (20 mM HEPES 7.5, 150 mM NaCl, 1 mM MgCl₂, 2 mM DTT, and 20 μM GDP) and incubated with 5X stocks of cyclic peptides at room temperature for 2 h. GTPγS binding was initiated by mixing with the reaction buffer at room temperature (50 nM [³⁵S]GTPγS and 100 μM GTPγS in dilution buffer) at room temperature. Final [cyclic peptide]: 10 μM; Final [Gα]: 2 μM; Final [GTPγS]: 100 μM. At various time points, 10 μL of the sample was removed and mixed with 390 μL of ice-cold wash buffer (20 mM HEPES 7.5, 150 mM NaCl, 20 mM MgCl₂). The mixture was filtered through a mixed cellulose membrane (25 mm, 0.22 μm). The membrane was washed by ice-cold wash buffer (500 μL x 3), put in a 6-mL plastic vial and air-dried (room temperature 1.5 h). 5 mL of CytoScint-ES Liquid Scintillation Cocktail (MP Biomedicals) was added to each vial. After

incubation overnight at room temperature, the vial was used for liquid scintillation counting with an LS 6500 Multi-Purpose Scintillation Counter. A standard curve was generated using [³⁵S]GTP γ S. The radioactive activity (Counts per minute) of the samples were converted to the GTP γ S concentration. The GTP γ S binding curves were fitted by the software GraphPad Prism using the following equation to calculate the apparent GTP γ S binding rates (k_{app}):

$$Y = \text{Plateau} * (1 - \exp(-k_{app} * X))$$

in which “Y” is the concentration of GTP γ S that bound to G α protein at time “X” (minutes).

FRET based G α /G $\beta\gamma$ interaction assay

Biotinylated avi-G α s (6-end, WT) and avi-G α i (FL, WT) were diluted to 32 nM (8X) using assay buffer (1X PBS 7.4, 2 mM DTT, 0.1% BSA, 2 mM MgCl₂, 0.05% Tween plus 0.5 mM GDP), followed by mixing with a same volume of 8X streptavidin XL665 stock (32 nM in the assay buffer). 8X His-G $\beta\gamma$ (C68S) stock (16 nM) and 8X anti-6His-Tb cryptate stock (0.4 μ g/mL) were added into the G α /XL665 mixtures. Finally, 2X stocks of cyclic peptides were (Final cyclic peptide concentrations were indicated in Figures) added with the protein mixtures. After incubation at room temperature for 2 h at room temperature. The plate was read on a TECAN Spark 20 M plate reader using the TR-FRET mode with the following parameters: Lag time: 70 μ s, Integration time: 500 μ s, Read A: Ex 320(25) nm (filter), Em 610(20) nm (filter), Gain 130, Read B: Ex 320(25) nm (filter), Em 665(8) nm (filter), Gain 165. FRET Signal was calculated as the ratio of [Read B]/[Read A]. Final [cyclic peptide]: 0, 0.002, 0.006, 0.019, 0.056, 0.169, 0.508, 1.524, 4.57, 12.7, 41.2, 123, 370, 1111, 3333, 10,000 nM; Final [G α]: 4 nM; Final [G β 1/ γ 2(C68S)]: 2 nM.

Crystallization

GN13/GNP/G α s complex: Wild type G α s (residues 7–380) that was preloaded with GNP and purified by gel filtration was concentrated to 10 mg/mL. The protein was then mixed with 1 mM of GNP (50 mM stock in H₂O) and 0.42 mM of the cyclic peptide GN13 (14 mM stock in DMSO). For crystallization, 0.2 μ L of the protein sample was mixed with 0.2 μ L of the well buffer containing 0.1 M HEPES 7.2, 20% PEG4000, 10% v/v 2-propanol. Crystals were grown at 20°C in a 96-well plate using the hanging-drop vapor-diffusion method, transferred to a cryoprotectant solution (0.1 M HEPES 7.2, 20% PEG4000, 10% v/v 2-propanol, 150 mM NaCl, 20 mM HEPES 8.0, 5 mM MgCl₂, 1 mM GNP, 25% v/v glycerol), and flash-frozen in liquid nitrogen.

GD20/GDP/G α s complex: Wild type G α s (NCBI Reference Sequence: NP_536351.1, residues 35–380) was preloaded with GDP, purified by gel filtration and then concentrated to 11.6 mg/mL. Before crystallization, the protein was mixed with 5 mM of Dithiothreitol (0.5 M stock in H₂O), 1 mM of GDP (50 mM stock in H₂O) and 0.76 mM of the cyclic peptide GD20 (42.6 mM stock in DMSO). For crystallization, 1.5 μ L of the protein sample was mixed with 1.5 μ L of the well buffer containing 0.1 M Tris 8.2, 26% PEG4000, 0.8 M LiCl. Crystals were grown at 20°C in a 15-well plate using the hanging-drop vapor-diffusion method, and flash-frozen in liquid nitrogen.

Data collection and structure determination

The dataset was collected at the Advanced Light Source beamline 8.2.1 with X-ray at a wavelength of 0.999965 Å. Then the dataset was integrated using the HKL2000 package (Otwinowski and Minor, 1997), scaled with Scala (Evans, 2006) and solved by molecular replacement using Phaser (McCoy et al., 2007) in CCP4 software suite (Winn et al., 2011). The crystal structure of GDP-bound human G α s R201C/C237 mutant (PDB code: 6AU6) was used as the initial model. The structure was manually refined with Coot (Emsley et al., 2010) and PHENIX (Adams et al., 2010). Data collection and refinement statistics are shown in Tables S1 and S2.

Chloroalkane penetration assay (CAPA)

The cell lines used for CAPA were HeLa cell lines, generated by Chenoweth and co-workers, that stably express HaloTag exclusively in the cytosol (Peraro et al., 2018). Cells were seeded in a 96-well plate the day before the experiment at a density of 4×10^4 cells per well. The day of the experiment the media was aspirated, and 100 μ L of cyclic peptide dilutions in DMEM were added to the cells. Plate was incubated for 19.5 h at 37°C with 5% CO₂. The contents of the wells were aspirated off, and wells were washed using fresh Opti-MEM for 15 min. The wash was aspirated off, and the cells were chased using 5 μ M ct-TAMRA for 15 min, except for the No-ct-TAMRA control wells, which were incubated with Opti-MEM alone. The contents of the wells were aspirated and washed with fresh Opti-MEM for 30 min. After aspiration, cells were rinsed once with PBS (PBS). The cells were then trypsinized, quenched with DMEM, resuspended in PBS, and analyzed using a benchtop flow cytometer (CytoFLEX, Beckman). Final [cyclic peptide]: 0, 0.034, 0.10, 0.31, 0.93, 2.78, 8.33, 25 μ M.

BRET2 based G α G $\beta\gamma$ interaction assay

The plasmids encoding M2R was a gift from Dr. Roderick MacKinnon. The plasmids encoding G α -RLuc8, G β 1, and GFP2-G γ 1 were gifts from Dr. Bryan Roth. The plasmid encoding GFP2-G γ 2 was generated by replacing the G γ 1 sequence of pcDNA3.1-GGgamma1-GFP2 by digestion with BamHI/XbaI and subsequent insertion of the G γ 2 sequence(MASNNTASIAQARKLVEQLKMEANIDRIKV-SKAAADLMAYCEAHAKEDPLLTVPASENPFREKKFFCAIL). All plasmids were sequenced to ensure their identities.

The BRET2 assay was conducted as reported (Olsen et al., 2020). Cells were plated in 10 cm dishes at 2.5–3 million cells per dish the night before transfection. Cells were transfected using a 6:6:3:1 DNA ratio of receptor:G α -RLuc8:G β :GFP2-G γ (750:750:375:125 ng for 10 cm dishes). Transit 2020 was used to complex the DNA at a ratio of 3 μ L Transit per μ g DNA, in

OptiMEM at a concentration of 10 ng DNA per μL OptiMEM. 16 h after transfection, cells were harvested from the plate using TrypLE and plated in poly-D-lysine-coated white, clear-bottom 96-well assay plates at a density of 30,000–35,000 cells per well.

8 h after plating in 96-well assay plates, media was replaced with 100 μL of cyclic peptide dilutions (Final cyclic peptide concentrations were indicated in Figures) in DMEM with 1% dialyzed FBS. 16 h after drug treatment at 37°C with 5% CO_2 , white backings were applied to the plate bottoms, and growth medium was carefully aspirated and replaced immediately with 60 μL of 1.67X drug dilutions in assay buffer (1 $\times$ Hank's balanced salt solution (HBSS) + 20 mM HEPES, pH 7.4), followed by a 10 μL addition of freshly prepared 50 μM coelenterazine 400a. After a 5 min equilibration period, cells were treated with 30 μL of 3.33X GPCR agonist or DMSO dilutions in assay buffer for an additional 5 min. Plates were then read in a TECAN Spark 20M plate reader with 395 nm (RLuc8-coelenterazine 400a) and 510 nm (GFP2) emission filters, at integration times of 1 s per well. Plates were read serially six times, and measurements from the fourth read were used in all analyses. BRET2 ratios were computed as the ratio of the GFP2 emission to RLuc8 emission.

Whole-cell voltage-clamp recordings

The plasmids encoding G β 1-C Venus, G γ 2-N Venus, and GIRK4-NLuc were gifts from Dr. Roderick MacKinnon. Cells were plated in 6 well plate at 0.55 million cells per well the night before transfection. Cells were transfected β 2AR (100 ng), G β 1-C Venus (25 ng), G γ 2-N Venus (25 ng), GIRK4-NLuc (100 ng). 1.75 μL of Lipofectamine (2000) was used to complex the DNA in 88 μL of OptiMEM. Transfected cells were incubated at 37°C for 12 h. After 12 h, cells were plated on glass coverslips and incubated at 37°C for 12 h for electrophysiological recordings. Whole-cell voltage-clamp recordings were performed with an Axopatch 200B amplifier (Molecular Devices, San Jose, CA) in the whole-cell mode. The analog current signal was low-pass filtered at 5 kHz (Bessel) and digitized at 50 kHz with a Digidata 1550B digitizer (Molecular Devices, San Jose, CA). Digitized data was recorded using the software pClamp 10.7. Patch electrodes (resistance 2.0–4.0 M Ω) were pulled on a Sutter P-97 puller (Sutter Instrument Company, Novato, CA) from 1.5 mm outer diameter filamented borosilicate glass. Extracellular solution contained 140 mM NaCl, 5 mM KCl, 2 mM CaCl_2 , 2 mM MgCl_2 , 10 mM D-glucose, 10 mM HEPES-NaOH (pH 7.4) ($\sim$ 330 mOsm). The extracellular solution was exchanged to high K^+ solution containing 40 mM NaCl, 100 mM KCl, 2 mM CaCl_2 , 2 mM MgCl_2 , 10 mM D-glucose, 10 mM HEPES-NaOH (pH 7.4) ($\sim$ 330 mOsm). The pipette solution contained 13.5 mM NaCl, 140 mM K-aspartate, 1.6 mM MgCl_2 , 0.09 mM EGTA-K, 9 mM HEPES-KOH (pH 7.2) ($\sim$ 290 mOsm). 1% DMSO, 25 μM cpGD20 in 1% DMSO, or 25 μM cpGD20-F5A in 1% DMSO was added to the pipette solution before the experiments.

Chemical stability assay

These assays were conducted by Pharmaron Beijing CO., Ltd. Cyclic peptides working solutions were prepared at 10 μM in DMEM with 10% FBS (Avantor, Cat# 76,294-180) or human plasma (Pooled, Male & Female, BioIVT, Cat# HMN666664). The assays were performed in duplicate. Vials were incubated at 37 °C at 60 rpm in a water bath and taken at designated time points including 0, 480, 1080 and 1440 min. For each time point, the initiation of the reaction was staggered so all the time points were terminated with cold acetonitrile containing internal standards (IS, 100 nM alprazolam, 200 nM labetalol, 200 nM Imipramine and 2 μM ketopifen) at the same time. Samples were vortexed then centrifuged at 4°C to remove proteins. The supernatants from centrifugation were diluted by ultra-pure H_2O and used for LC-MS/MS analysis. All calculations were carried out using GraphPad Prism. Remaining percentages of parent compounds at each time point were estimated by determining the peak area ratios from extracted ion chromatograms.

Chemical synthesis

Solid phase synthesis of cyclic peptides: Macrocyclic peptides (25 μmol scale) were synthesized by a standard Fmoc solid phase peptide synthesis method using a Syro Wave automated peptide synthesizer (Morimoto et al., 2012). After addition of a chloroacetyl group onto the N-terminal amide group (for the formation of cyclic peptide), peptides were cleaved from the NovaPEG Rink Amide resin (Novabiochem) by a solution of 92.5% trifluoroacetic acid (TFA), 2.5% 3,6-Dioxo-1,8-octanedithiol ethanedithiol (DOTD), 2.5% triisopropylsilane (TIPS) and 2.5% water and precipitated by diethyl ether. To conduct the macrocyclization reaction, the peptide pellet was dissolved in 10 mL DMSO containing 10 mM tris(2-carboxyethyl)phosphine hydrochloride (TCEP), adjusted to pH > 8 by addition of triethylamine (TEA) and incubated at 25°C for 1 h. This cyclization reaction was quenched by acidification of the solution with TFA. The crude products were purified by reverse-phase HPLC (RP-HPLC) (Shimadzu) with a Chromolith RP-18 100-25 prep column. Molecular masses were verified by a time-of-flight mass spectrometer (Waters Xevo G2-XS), and the purity was verified by analytical HPLC on a Waters Acquity UPLC BEH C18 1.7 μm column.

General synthesis route of chloroalkane tagged cyclic peptides: In this work, we prepared a chloroalkane tag (ct) that has been previously used with the HaloTag system (Neklesa et al., 2011). Instead of using the Rink amide resin, peptides were synthesized using the Fmoc-Wang resin (Anaspec, AS-20058) to generate a carboxylate at the C-terminus. To cap the C-terminus with the chloroalkane tag (ct), 10 equiv of chloroalkane tag (ct), 5 equiv of HATU, and 20 equiv of DIPEA were dissolved in DMF and stirred for 1 h at room temperature. Crude peptides were purified by reverse-phase HPLC (Waters XBridge C18 column 5 μm particle size 30 $\times$ 250 mm, 5%–95% acetonitrile-water + 0.1% formic acid, 40 min, 20 mL/min) to afford the chloroalkane tagged peptides.

Characterization data for cyclic peptides

Mass spectrometry

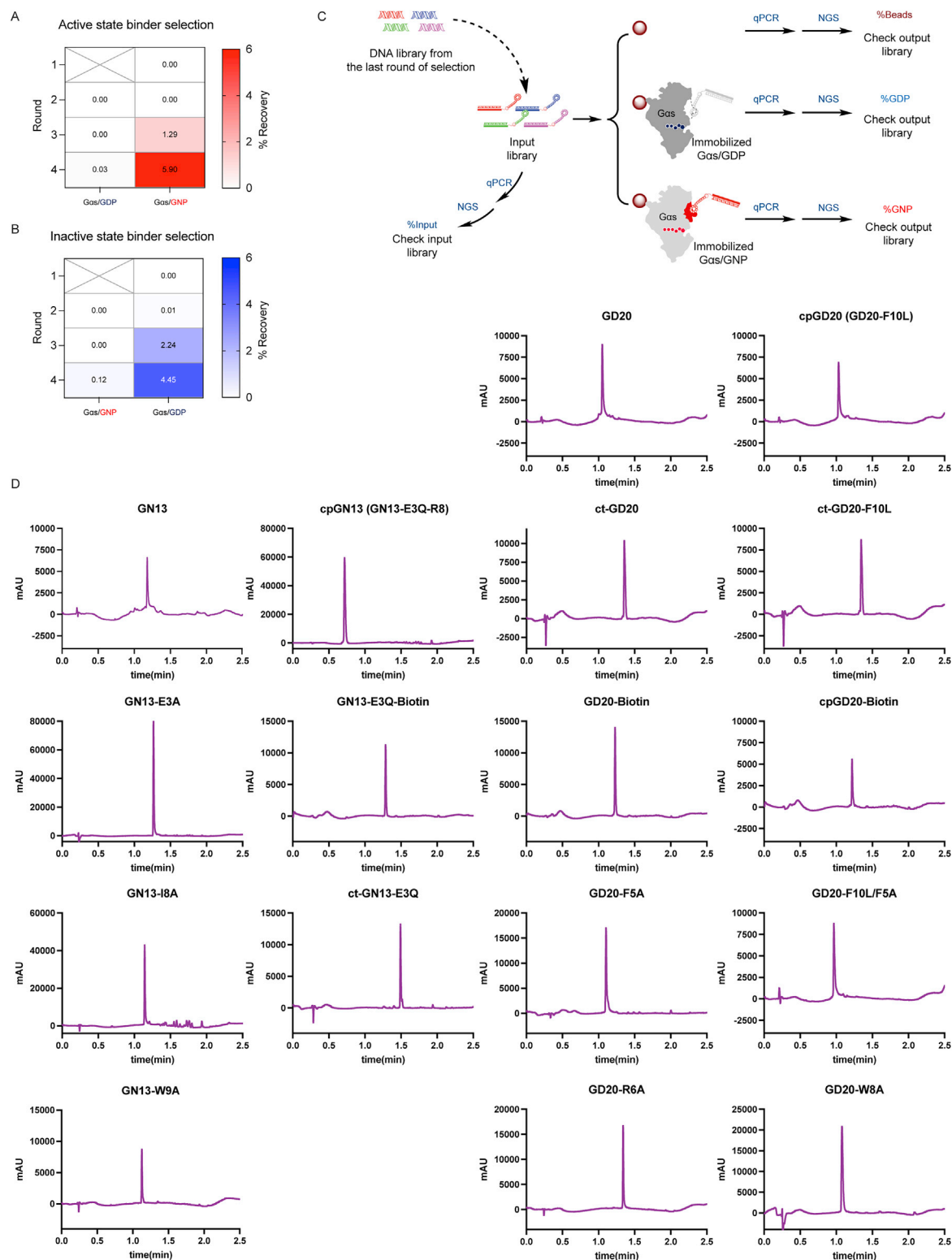
GN13: HRMS (ESI): Calcd for $(C_{79}H_{106}N_{16}O_{21}S + 2H)^{2+}$: 824.3798, Found: 824.3973.
 GN13-E3A: HRMS (ESI): Calcd for $(C_{77}H_{104}N_{16}O_{19}S + 2H)^{2+}$: 795.3770, Found: 795.3749.
 GN13-I8A: HRMS (ESI): Calcd for $(C_{76}H_{100}N_{16}O_{21}S + 2H)^{2+}$: 803.3563, Found: 803.3563.
 GN13-W9A: HRMS (ESI): Calcd for $(C_{71}H_{101}N_{15}O_{21}S + 2H)^{2+}$: 766.8587, Found: 766.8610.
 cpGN13: HRMS (ESI): Calcd for $(C_{140}H_{223}N_{53}O_{37}S + 3H)^{3+}$: 1091.2384, Found: 1091.5806.
 GN13-E3Q-Biotin: HRMS (ESI): Calcd for $(C_{113}H_{170}N_{20}O_{33}S_2 + 2H)^{2+}$: 1200.5919, Found: 1200.5970.
 ct-GN13-E3Q: HRMS (ESI): Calcd for $(C_{89}H_{126}ClN_{17}O_{22}S + 2H)^{2+}$: 926.9415, Found: 926.9422.
 GD20: HRMS (ESI): Calcd for $(C_{90}H_{126}N_{22}O_{20}S + 2H)^{2+}$: 934.4698, Found: 934.4844.
 cpGD20 (GD20-F10L): HRMS (ESI): Calcd for $(C_{87}H_{128}N_{22}O_{20}S + 2H)^{2+}$: 917.4776, Found: 917.4901.
 ct-GD20: HRMS (ESI): Calcd for $(C_{100}H_{145}ClN_{22}O_{22}S + 2H)^{2+}$: 1037.5235, Found: 1037.5303.
 ct-GD20-F10L: HRMS (ESI): Calcd for $(C_{97}H_{147}ClN_{22}O_{22}S + 2H)^{2+}$: 1020.5313, Found: 1020.5193.
 GD20-Biotin: HRMS (ESI): Calcd for $(C_{124}H_{189}N_{25}O_{33}S_2 + 2H)^{2+}$: 1311.1739, Found: 1311.1741.
 cpGD20-Biotin: HRMS (ESI): Calcd for $(C_{121}H_{191}N_{25}O_{33}S_2 + 2H)^{2+}$: 1294.1817, Found: 1294.1805.
 GD20-F5A: HRMS (ESI): Calcd for $(C_{84}H_{122}N_{22}O_{20}S + 2H)^{2+}$: 896.4542, Found: 896.4604.
 cpGD20-F5A: HRMS (ESI): Calcd for $(C_{81}H_{124}N_{22}O_{20}S + 2H)^{2+}$: 879.4620, Found: 879.4648.
 GD20-R6A: HRMS (ESI): Calcd for $(C_{87}H_{119}N_{19}O_{20}S + 2H)^{2+}$: 891.9378, Found: 891.9394.
 GD20-W8A: HRMS (ESI): Calcd for $(C_{82}H_{121}N_{21}O_{20}S + 2H)^{2+}$: 876.9487, Found: 876.9509

Absorbance was recorded at 280 nm (Figure S1D).

QUANTIFICATION AND STATISTICAL ANALYSIS

All of the curves in Figures except those from the BLI experiments were fitted by GraphPad Prism. Raw kinetic data collected from the BLI experiments were processed with the Data Analysis software provided by the manufacturer. All the details can be found in the figure legends and in the Method details. The data collection and refinement statistics of the crystal structures can be found in Tables S1 and S2 (related to Figures 3 and 5, see also Figures S3 and S5).

Supplemental figures



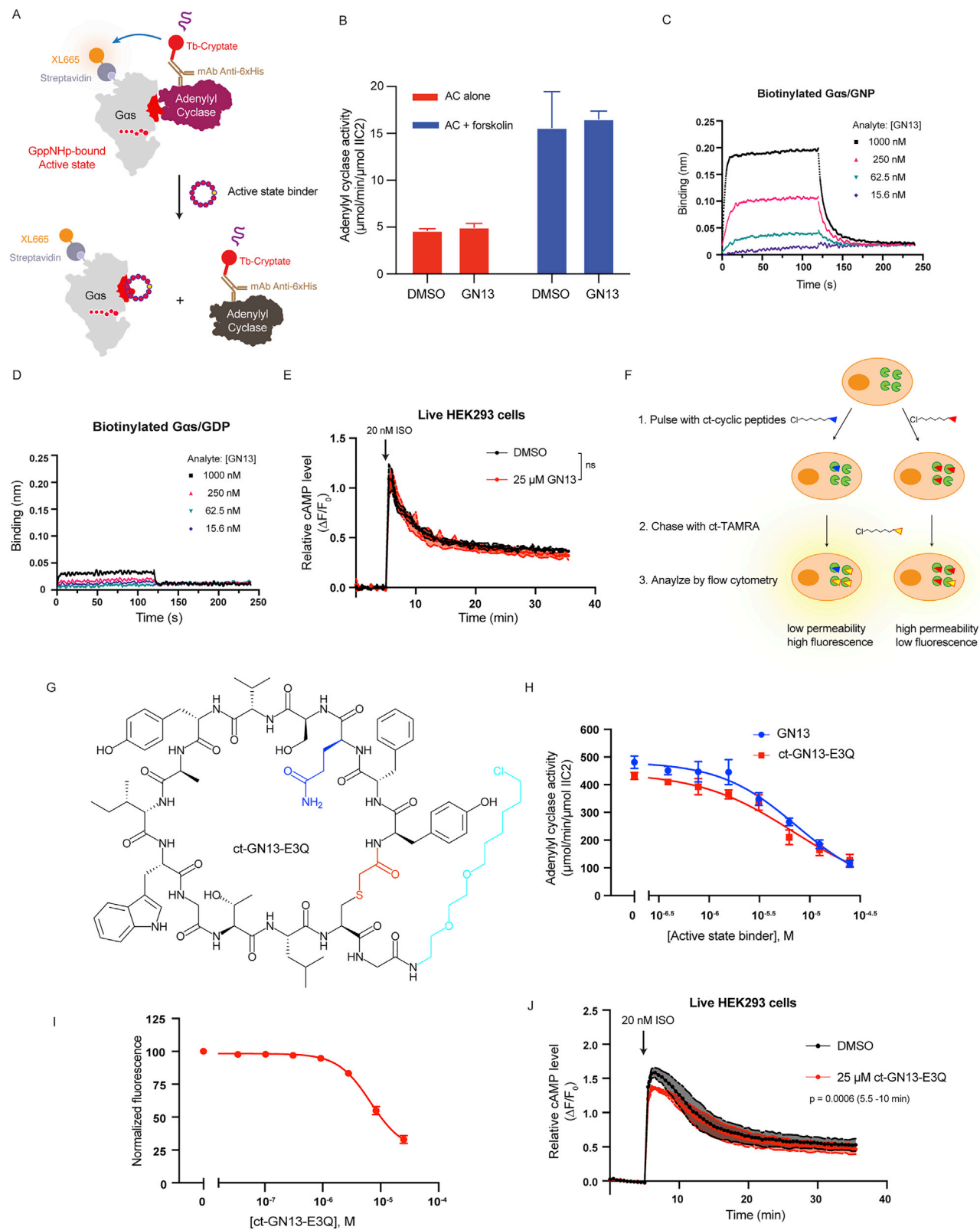
(legend on next page)

Figure S1. RaPID selection of state-selective G α s binding cyclic peptides, related to Figure 1

(A and B) The percentage of enriched peptide-mRNA-cDNA complex in the input library after each selection was quantified by qPCR. Cyclic peptides that bind to GNP-bound (A) or GDP-bound (B) G α s were enriched through R1-R4. To ensure a maximum library diversity at the initial stage of selection, negative selection was not included in the first round of selection.

(C) Comparison selection. DNA sequences of cyclic peptide binders from the R4 pools were quantified and identified by qPCR and NGS. A peptide-mRNA-cDNA complex library was produced based on the above-mentioned DNA sequences and equally split into three fractions. Binding of each individual peptide-mRNA-cDNA complex to blank, GDP-bound G α s-immobilized or GNP-bound G α s-immobilized beads was quantified by qPCR and NGS, respectively.

(D) Analytical HPLC Traces of resynthesized cyclic peptides. Absorbance was recorded at 280 nm.



(legend on next page)

Figure S2. $G\alpha s$ active-state inhibitor GN13 inhibits $G\alpha s$ -mediated adenylyl cyclase activation, related to Figure 2

(A) Illustration of active-state binders inhibiting PPI between $G\alpha s$ /GNP and AC. (B) GN13 did not directly inhibit the intrinsic or forskolin-mediated AC activity in the absence of $G\alpha s$. Mean $\pm$ SE, $n = 3$.

(C and D) Binding kinetics of GN13 to GNP-bound (C) or GDP-bound (D) $G\alpha s$ were quantified using bio-layer Interferometry (BLI). Biotinylated $G\alpha s$ proteins were immobilized to give a relative intensity of 2–3 nm on streptavidin biosensors. Association ($t = 0$ –120 s) and dissociation ($t = 120$ –240 s) cycles of compounds were started by dipping sensors into cyclic peptide solutions and control buffer. Binding signals were reference-subtracted. The assay was performed in duplicate, and the data represent one of the two replicates.

(E) Pretreatment with GN13 for 24 h did not inhibit ISO-stimulated cAMP production in live HEK293 cells. Mean $\pm$ SD, $n = 3$. Two-tailed unpaired t tests (data after 5 min). ns $p > 0.05$.

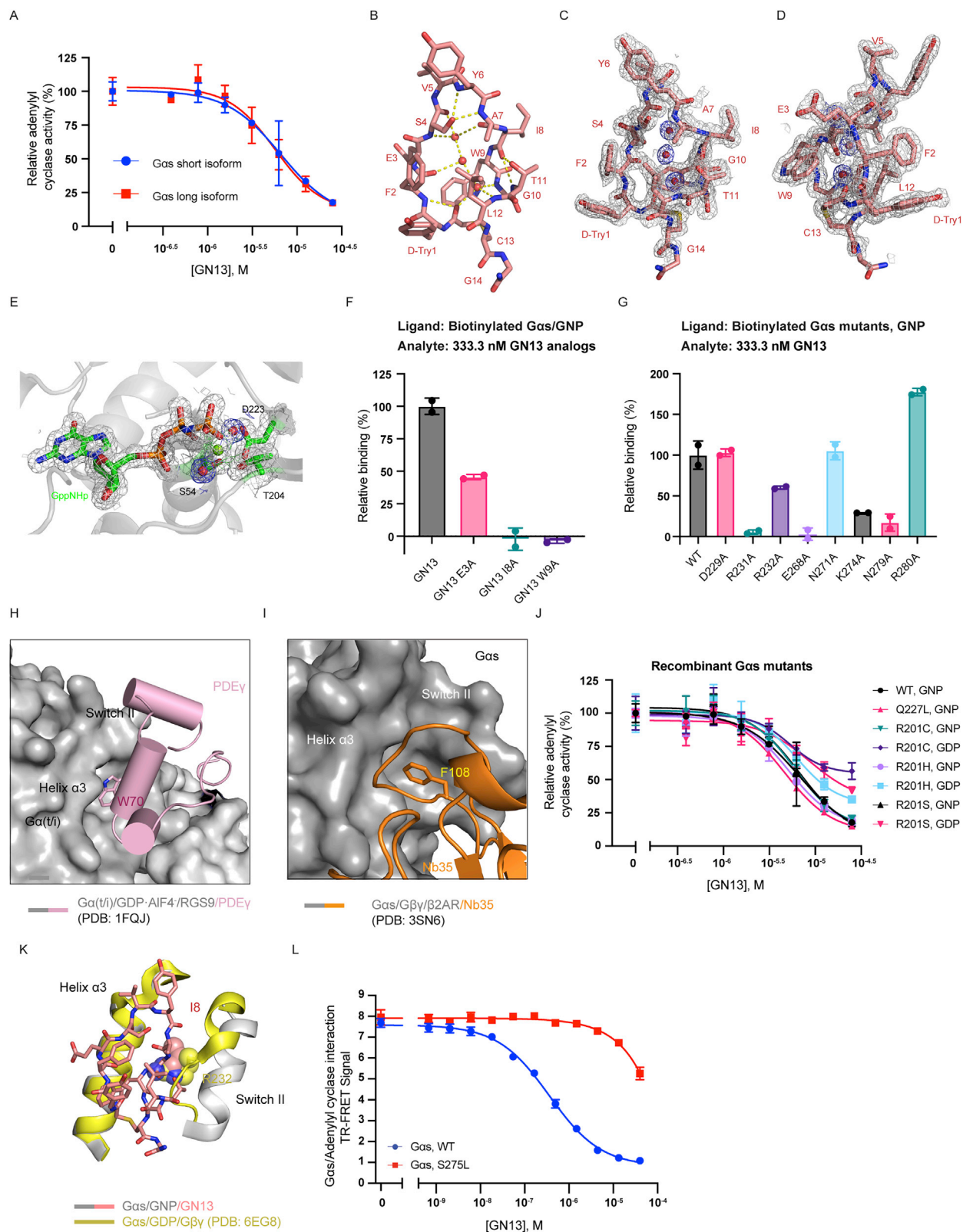
(F) Illustration of the chloroalkane penetration assay (CAPA).

(G) Structure of ct-GN13-E3Q. E3Q is colored blue. The ct tag is colored cyan.

(H) Activation of AC by $G\alpha s$ was inhibited by both GN13 and ct-GN13-E3Q in a dose-dependent manner. Mean $\pm$ SE, $n = 3$.

(I) CAPA cell permeability assay result of ct-GN13-E3Q. Mean $\pm$ SD, $n = 3$.

(J) Pretreatment with ct-GN13-E3Q for 24 h slightly inhibited ISO-stimulated cAMP production in live HEK293 cells. Mean $\pm$ SD, $n = 3$. Two-tailed unpaired t tests (data between 5.5 and 10 min), $p < 0.05$ was considered significant.



(legend on next page)

Figure S3. GN13 specifically inhibits G α s through binding to a crystallographically defined pocket, related to Figure 3

- (A) Activation of AC by both short and long isoforms of G α s was inhibited by GN13. AC activity was normalized to DMSO control group (100%). Mean $\pm$ SD, n = 3.
- (B) GN13 adopts a highly ordered three-dimensional structure through an H-bond network. GN13 is shown as salmon sticks. Three water molecules with well-defined electron density are shown as red spheres. H-bonds are represented by yellow dash lines.
- (C and D) Electron density map of GN13. The *2mFo-DFc* electron density map of the structure is contoured at 1.0 σ and colored gray (GN13) and blue (Water), respectively.
- (E) Electron density map of GNP. GNP and the side chains of S54, T204 and D223 are shown as sticks. The Mg²⁺ and two water molecules coordinated with the Mg²⁺ are shown as green and red spheres, respectively. The *2mFo-DFc* electron density map of the structure is contoured at 1.0 σ .
- (F) Binding of GN13 analogs to WT G α s/GNP were quantified using BLI. Biotinylated WT G α s/GNP was immobilized to give a relative intensity of 2–3 nm on streptavidin biosensors, following the same association/dissociation cycles described in Figure S2C. Binding signals were double referenced and normalized to G α s loading and GN13/G α s binding signal. Mean $\pm$ SD, n = 2.
- (G) Binding of GN13 to different GNP-bound G α s mutants were quantified using BLI. Biotinylated GNP-bound G α s proteins were immobilized to give a relative intensity of 2–3 nm on streptavidin biosensors, following the same association/dissociation cycles described in Figure S2C. Binding signals were double referenced and normalized to G α s loading and GN13/WT G α s binding signal. Mean $\pm$ SD, n = 2.
- (H) Structure of the GDP•AIF4⁻-bound G α (t/i)/RGS9/PDE γ complex (PDB: 1FQJ). A critical tryptophan residue from PDE γ (pink, cartoon) engages the hydrophobic pocket between the switch II region and the α 3 helix. G α (t/i) and RGS9 are shown as surface. PDE γ is shown as cartoon.
- (I) Structure of the G α s/G $\beta\gamma$ / β 2AR/Nb35 complex (PDB: 3SN6). A critical phenylalanine residue from Nb35 (orange, cartoon) engages the hydrophobic pocket between the switch II region and the α 3 helix. G α s is shown as surface. Nb35 is shown as cartoon.
- (J) Activation of AC by G α s oncogenic mutants was inhibited by GN13. AC activity was normalized to DMSO control group (100%). Mean $\pm$ SD, n = 3.
- (K) Structural basis for nucleotide-state-selective binding of GN13 to G α s. In G α s/GDP (yellow), switch II is partially disordered, which disrupts polar contacts with GN13 and creates extensive steric hindrance. In particular, R232 of switch II (shown in space filling) is predicted to create a steric clash with I8 of GN13.
- (L) GN13 inhibited PPI between G α s WT and AC. This inhibitory effect was significantly diminished by the S275L mutation (red). Mean $\pm$ SD, n = 3.

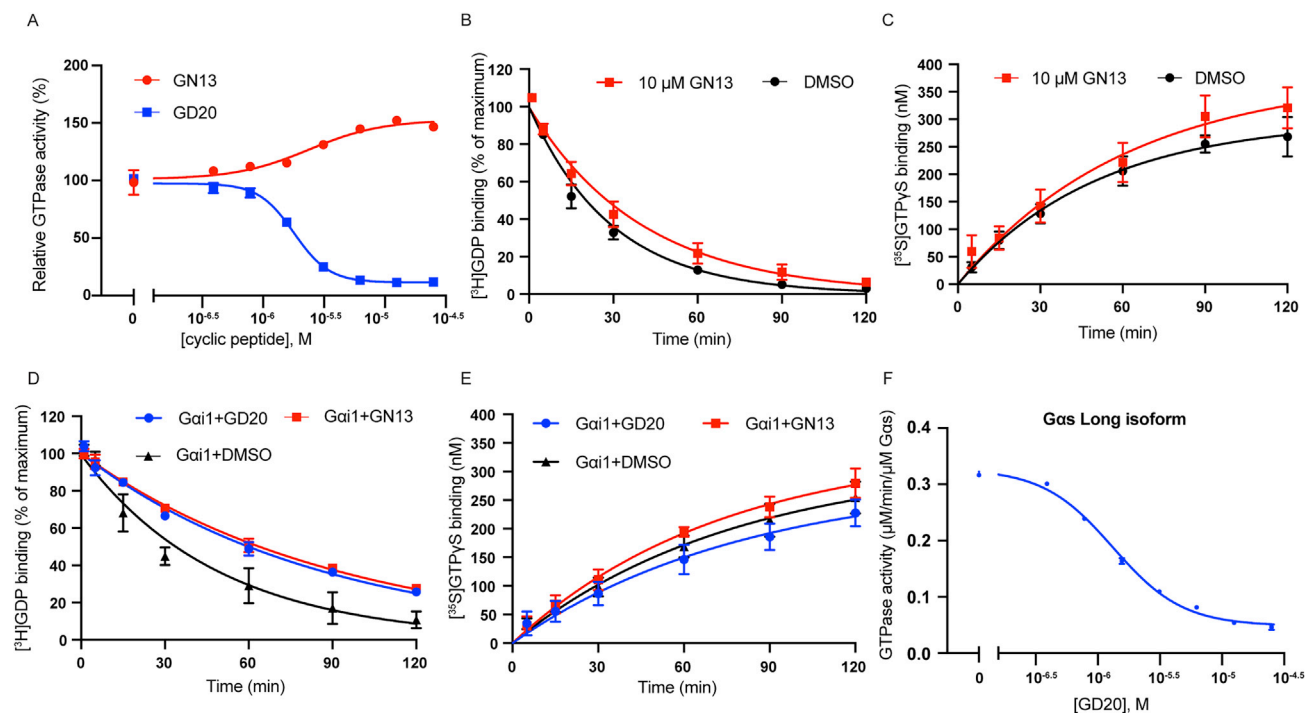
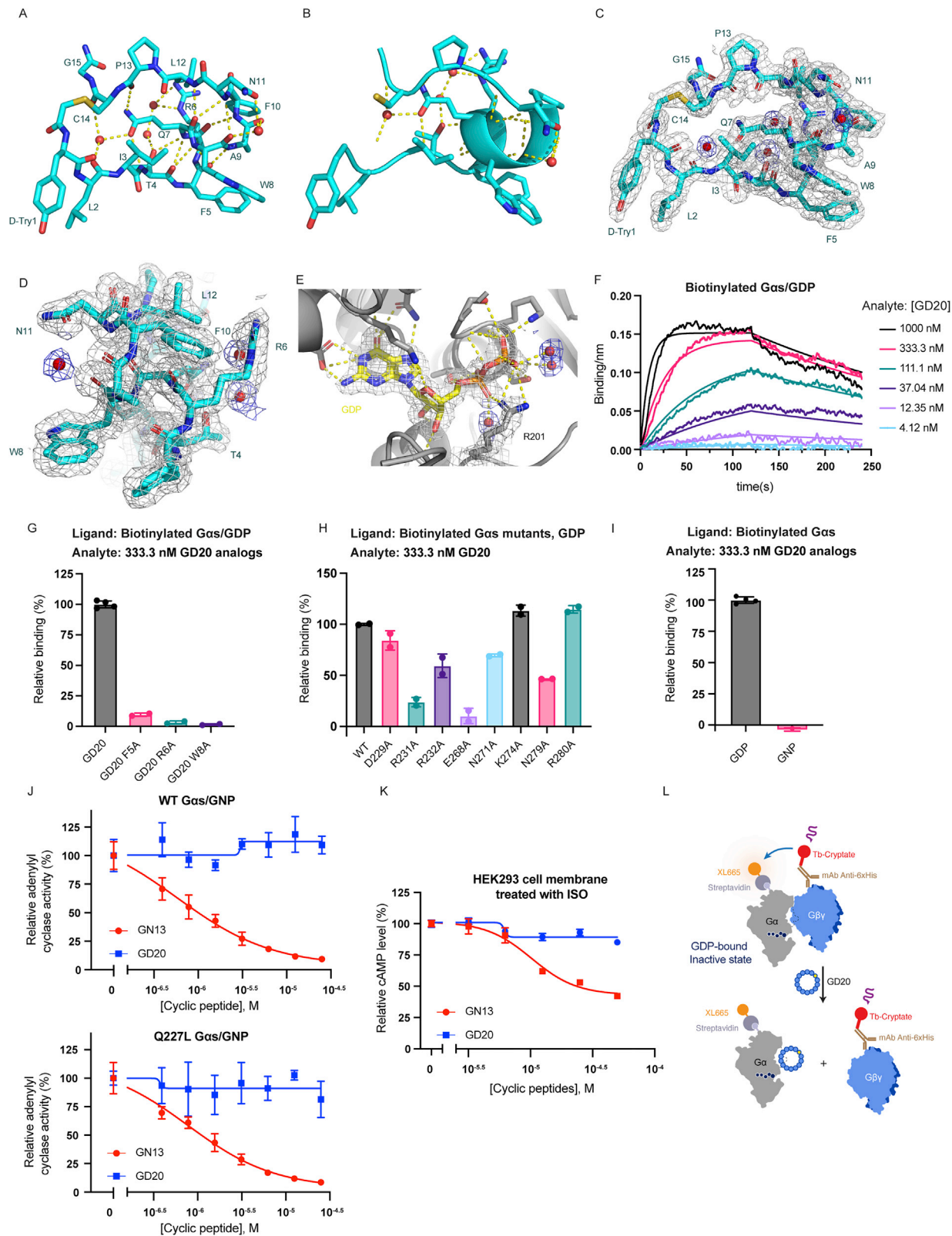


Figure S4. GN13 and GD20 modulate $G\alpha_s$ GTPase activity in a $G\alpha_s$ -specific manner, related to Figure 4

- (A) $G\alpha_s$ steady-state GTPase activity was modulated by GN13 and GD20. Mean $\pm$ SD, $n = 2$.
 (B) GDP dissociation from $G\alpha_s$ in the presence (red) or absence (black) of 10 μM GN13 were measured. Mean $\pm$ SD, $n = 3$.
 (C) GTP γ S binding to $G\alpha_s$ in the presence (red) or absence (black) of 10 μM GN13 were measured. Mean $\pm$ SD, $n = 3$.
 (D) GDP dissociation from Gai1 in the presence of 10 μM GN13 (red), or 10 μM GD20 (blue) or DMSO (black) were measured. Mean $\pm$ SD, $n = 3$.
 (E) GTP γ S binding to Gai1 in the presence of 10 μM GN13 (red), or 10 μM GD20 (blue) or DMSO (black) were measured. Mean $\pm$ SD, $n = 3$.
 (F) Steady-state GTPase activities of short and long isoforms of $G\alpha_s$ were inhibited by GD20. Mean $\pm$ SD, $n = 2$.



(legend on next page)

Figure S5. GD20 specifically inhibits G α s through binding to a crystallographically defined pocket, related to Figure 5

(A and B) GD20 adopts a highly ordered three-dimensional structure through an H-bond network. GD20 is shown as cyan sticks (A) or cartoon (B). Four water molecules with well-defined electron density are shown as red spheres. H-bonds are represented by yellow dash lines.

(C and D) Electron density map of GD20. The *2mFo-DFc* electron density map of the structure is contoured at 1.0 σ and colored gray (GD20) and blue (Water), respectively.

(E) Electron density map of GDP. GDP and the side chain of R201 are shown as sticks. The Mg²⁺ and two water molecules coordinated with the Mg²⁺ are shown as green and red spheres, respectively. The *2mFo-DFc* electron density map of the structure is contoured at 1.0 σ .

(F) Binding kinetics of GD20 to WT G α s/GDP were quantified using BLI. Biotinylated WT G α s/GDP was immobilized to give a relative intensity of 2–3 nm on streptavidin biosensors, following the same association/dissociation cycles described in Figure S2C. Binding signals were reference-subtracted. The assay was performed in duplicate, and the data represent one of the two replicates.

(G) Binding of GD20 analogs to WT G α s/GDP were quantified using BLI. Biotinylated WT GDP-bound G α s proteins were immobilized to give a relative intensity of 2–3 nm on streptavidin biosensors, following the same association/dissociation cycles described in Figure S2C. Binding signals were double referenced and normalized to G α s loading and GD20/G α s binding signal. Mean $\pm$ SD, n = 2.

(H) Binding of GD20 to different GDP-bound G α s mutants were quantified using BLI. Biotinylated GDP-bound G α s proteins were immobilized to give a relative intensity of 2–3 nm on streptavidin biosensors, following the same association/dissociation cycles described in Figure S2C. Binding signals were double referenced and normalized to G α s loading and GDP/WT G α s binding signal. Mean $\pm$ SD, n = 2.

(I) Binding of GD20 analogs to WT GDP-bound or GNP-bound G α s were quantified using BLI. Biotinylated WT G α s proteins were immobilized to give a relative intensity of 2–3 nm on streptavidin biosensors, following the same association/dissociation cycles described in Figure S2C. Binding signals were double referenced and normalized to G α s loading and GD20/GDP-bound G α s binding signal. Mean $\pm$ SD, n = 2.

(J) Activation of AC by GNP-bound WT G α s or GNP-bound G α s oncogenic mutant Q227L were inhibited by GN13 but not GD20. AC activity was normalized to DMSO control group (100%). Mean $\pm$ SD, n = 3.

(K) GD20 did not inhibit ISO-stimulated G α s activation in HEK293 cell membranes. Mean $\pm$ SD, n = 3.

(L) Illustration of inactive-state binders inhibiting PPI between G α s/GDP and G $\beta\gamma$ (C68S).

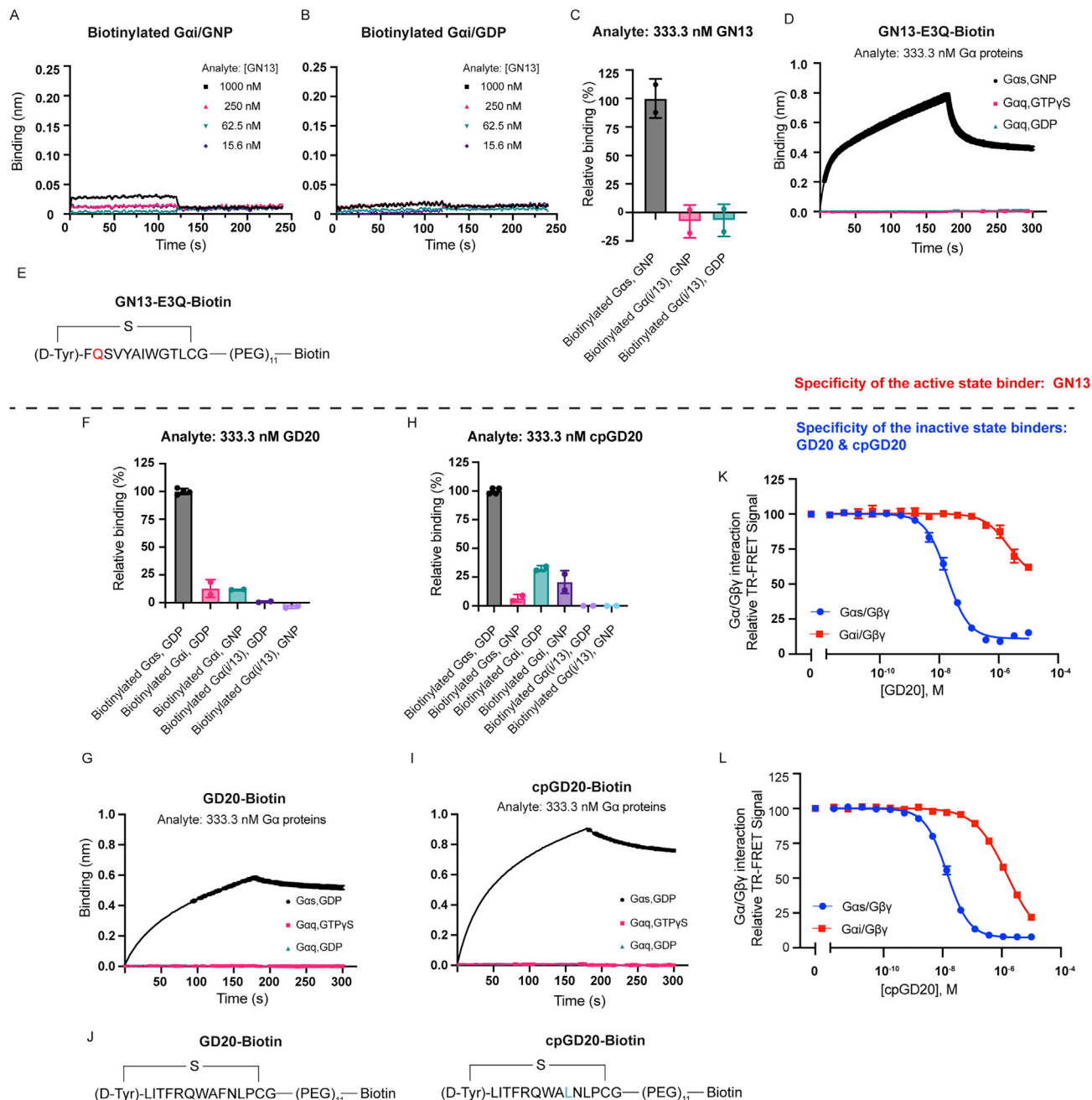


Figure S6. G protein class-specificity of GN13 and GD20, related to Figure 6

(A and B) Binding kinetics of GN13 to Gαi were quantified using BLI. Biotinylated Gαi proteins were immobilized to give a relative intensity of 2–3 nm on streptavidin biosensors, following the same association/dissociation cycles described in Figure S2C. Binding signals were reference-subtracted. The assay was performed in duplicate, and the data represent one of the two replicates.

(C) Binding of GN13 to Gαs and Gα(i/13) were quantified using BLI. The substitution of the N-terminal helix of Gαi1 for the corresponding region of Gα13 generated soluble chimeric Gα(i/13) protein. The N-terminal helix substitution is far away from GN13 binding interface, therefore will not influence its binding. Biotinylated Gα proteins were immobilized to give a relative intensity of 2–3 nm on streptavidin biosensors, following the same association/dissociation cycles described in Figure S2C. Binding signals were double referenced and normalized to Gα protein loading and GN13/GNP-bound Gαs binding signal. Mean ± SD, n = 2.

(D) The Avi tagged Gαq was insoluble (data not shown), therefore, biotinylated GN13-E3Q was immobilized to give a relative intensity of 0.2–0.3 nm on streptavidin biosensors. GN13-E3Q was chosen to simplify chemical synthesis of biotinylated cyclic peptides. Binding kinetics of untagged Gαs and Gαq to immobilized GN13-E3Q were quantified using BLI. Association (t = 0–180 s) and dissociation (t = 180–300 s) cycles of Gα proteins were started by dipping sensors into Gα protein solutions and control buffer. Binding signals were double referenced. Mean ± SD, n = 2.

(E) Design of biotinylated GN13-E3Q.

(legend continued on next page)

(F and H) Binding of GD20 (F) or cpGD20 (H) to $G_{\alpha s}$, $G_{\alpha i}/13$ and $G_{\alpha i}$ were quantified using BLI. Biotinylated G_{α} proteins were immobilized to give a relative intensity of 2–3 nm on streptavidin biosensors, following the same association/dissociation cycles described in Figure S2C. Binding signals were double referenced and normalized to G_{α} protein loading and the GD20(or cpGD20)/GDP-bound $G_{\alpha s}$ binding signal. Mean $\pm$ SD, $n = 2$.

(G and I) Biotinylated GD20 (G) or Biotinylated cpGD20 (I) was immobilized to give a relative intensity of 0.3–0.4 nm on streptavidin biosensors. Binding kinetics of untagged $G_{\alpha s}$ and $G_{\alpha q}$ to immobilized GD20 or cpGD20 were quantified using BLI. Association ($t = 0$ –180 s) and dissociation ($t = 180$ –300 s) cycles of G_{α} proteins were started by dipping sensors into G_{α} protein solutions and control buffer. Binding signals were double referenced. Mean $\pm$ SD, $n = 2$.

(J) Design of biotinylated GD20 and biotinylated cpGD20 (GD20-F10L).

(K) GD20 inhibited PPI between $G_{\alpha s}$ /GDP and $G\beta\gamma$ (C68S). GD20 was 100-fold more selective for $G_{\alpha s}$ than $G_{\alpha i}$. Mean $\pm$ SD, $n = 3$.

(L) cpGD20 inhibited PPI between $G_{\alpha s}$ /GDP and $G\beta\gamma$ (C68S). cpGD20 was nearly 100-fold more selective for $G_{\alpha s}$ than $G_{\alpha i}$. Mean $\pm$ SD, $n = 3$.

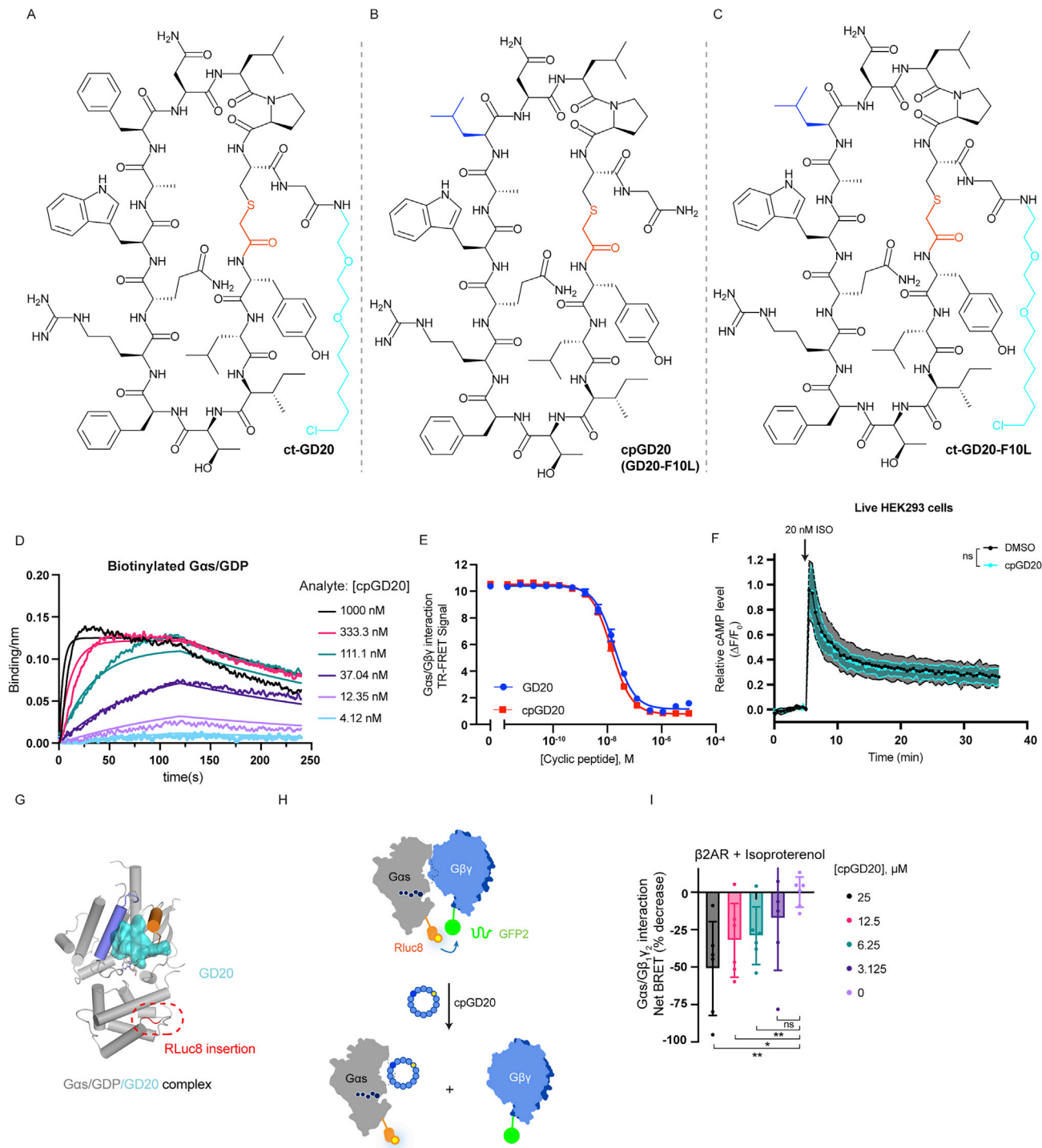


Figure S7. A cell-permeable GD20 analog, cpGD20, is a dual-effect G protein modulator, related to Figure 7

(A–C) Structure of derivatized cyclic peptides. (A) ct-GD20 (B) cpGD20 (GD20-F10L) (C) ct-GD20-F10L.

(D) Binding kinetics of cpGD20 to WT *Gas*/GDP were quantified using BLI. Biotinylated WT *Gas*/GDP was immobilized to give a relative intensity of 2–3 nm on streptavidin biosensors, following the same association/dissociation cycles described in Figure S2C. Binding signals were reference-subtracted. The assay was performed in duplicate, and the data represent one of the two replicates.

(E) cpGD20 inhibited PPI between *Gas*/GDP and *Gβγ*(C68S). Mean $\pm$ SD, $n = 3$.

(legend continued on next page)

(F) Pretreatment with cpGD20 for 24 h did not inhibited ISO-stimulated cAMP production in live HEK293 cells. Mean $\pm$ SD, n = 3. Two-tailed unpaired t tests (data after 5 min). ns p > 0.05.

(G) The GD20/*G α s* complex structure provides structural basis for the Rluc8 insertion. Rluc8 is inserted between α B and α C helices.

(H) Illustration of cpGD20 inhibiting PPI between *G α s*Short-Rluc and G β 1/GFP2- γ 2 in a BRET2 assay.

(I) cpGD20 inhibited *G α s*/G β γ reassociation in HEK293 cells in a dose dependent manner. *G α s*/G β γ dissociation was measured by BRET2 signal reduction after 1 nM ISO application. BRET signal was normalized to cells that were not treated with ISO and the percentage decrease was calculated based on the net BRET2 signal at [cpGD20] = 0 μ M. Mean $\pm$ SD, n = 6. Two-tailed unpaired t tests, *p < 0.05, **p < 0.01, ns p > 0.05.